

The Biological Effects of Microwaves and Related Questions

H. FRÖHLICH

*Department of Physics
University of Liverpool
Liverpool, England*

I. Introduction	85
II. General	87
A. Biology	87
B. Physics	89
III. Theory	92
A. Material	92
B. Metastable Highly Polar States	95
C. Excitation of Coherent Vibrations	98
IV. Consequences	104
A. General	104
B. Long-Range Interaction	106
C. Models for Nonthermal Actions of Microwaves	109
D. Multicomponent Systems and the Cancer Problem	113
E. Brain Waves	118
V. Experiments	121
A. General	121
B. Experiments with Millimeter Waves	127
C. Raman Effect	140
VI. Conclusions	148
References	150

I. INTRODUCTION

Microwaves, and particularly millimeter waves, are strongly absorbed in water, which represents a most important constituent of biological systems. As a consequence, when irradiated by microwaves the temperature of the biological material is raised and temperature-dependent processes are influenced. Well-documented biological effects exist, however, arising from irradiation at very low intensities where thermal effects are excluded. This article deals with these effects and with their prediction

from theoretical considerations. It must be emphasized at once that an appropriate discussion of experimental results with regard to nonthermal effects must frequently consider thermal effects as well. Much prejudice has arisen from an inadequate knowledge of the basic differences between thermal and nonthermal effects. Thus the amount of energy transfer from the radiation to the material varies slowly with frequency and is largely governed by the dielectric loss. Within reasonable limits this loss is proportional to the intensity of the radiation. Nonthermal effects, on the other hand, occur in certain frequency regions only, and usually exhibit saturation at rather low intensity. As a consequence, nonthermal effects might be drowned by thermal ones. This is in particular to be expected when the sign of the considered effect in the nonthermal region is opposite to that of the thermal effect. As an example, consider the reduction of the rate of growth of certain bacteria arising from irradiation at 73.6 Hz, as discussed in Section V. Temperature increase, on the other hand, increases the rate of growth, thus canceling the nonthermal effect at sufficiently high intensity.

It might be tempting to link frequency-dependent biological effects of a resonance type to absorption bands in certain biomolecules. We shall demonstrate, however, that such resonances are properties of the whole system, and may depend on the biological activity. It is of importance, in this context, to realize that in some instances biosystems can exhibit properties of the most refined electronic instruments, as discussed in Section II,A. They achieve this with the use of biomolecules in a way that is not yet understood.

How is it possible that extremely weak electromagnetic radiation can have large biological consequences? To answer this it must be realized that an active biological system is very far from thermal equilibrium and has considerable amounts of energy stored for appropriate use. The microwave might then be considered as a trigger for certain bioeffects. The electronic engineer might like the analogy of a radio receiver that under the influence of an extremely weak electromagnetic wave (from a sender) will produce large sound effects (corresponding to bioeffects) when appropriately tuned. Switched off, however, it would be very difficult, though not impossible, to observe resonances.

A great number of experiments have been published in which the influence of irradiation by microwaves of 2450 MHz—a frequency at which water absorbs strongly—on a variety of biological effects has been investigated. While not completely useless, this type of research does not lend itself to systematic discussion, which will require variation of frequency or of intensity—preferably of both.

In the interest of a systematic presentation, the theoretical (physical)

background is first presented on the basis of certain models. It must be recognized, however, that theoretical physics cannot make predictions in the way it does, e.g., in the realm of the linear response of more usual physical materials. Biological systems are in some respects very far from equilibrium, quite outside the regions of linear response. They have developed to fulfil a certain purpose, and usually there is no unique way in which this can be performed. Theoretical physics must, however, provide the relevant concepts, and it is here that the possibility of trigger action of microwaves arises. This holds in particular for the excitation of coherent electric vibrations pumped by energy derived from metabolism. These will be shown likely to have frequencies that can resonate with those of microwaves—although other frequencies are also likely to be relevant. For this reason it is not desirable to restrict our discussion to microwaves only.

Such excitations can have far-reaching consequences on biological activity. While theory can point out a number of possibilities, it cannot predict in which way evolution has progressed. It is only in collaboration with experiment that the possibilities that have actually been established can be found.

The attitude we thus must take is to consider biological systems in terms of their activities, which require a high degree of organization. Such organization may be of a complex nature, but it does require consideration of the systems as a whole. In microscopic terms it thus requires the introduction of collective properties that in some cases might go beyond those normally used in physics. It is sometimes asserted that experimental proof of the obedience of microscopic laws of physics and chemistry invalidates the existence of collective properties, but such conclusions are illogical. Thus a particular chemical process will be determined in terms of the atomic constitution of the systems undergoing a chemical transformation, i.e., will be determined microscopically. The fact, however, that the particular systems are brought together at a particular time in a particular place may well arise from certain collective properties of a larger region.

II. GENERAL

A. Biology

Frequently the opinion has been expressed that the activity of biological systems can be understood in terms of standard chemical reactions in conjunction with diffusion of molecules. On this basis the biological ef-

fects of weak microwaves would have to be considered as nonsense. For this reason it is necessary to point to a few biological properties that equally might have been considered as nonsense were they not so extremely well documented.

In the first place we note that the biological sensitivity to certain external influences can equal that of the best available instrumentation. This will be illustrated on two cases showing that high sensitivity to microwaves need not be an isolated feature.

In a careful analysis, Rose (1970) has demonstrated that at low light intensities the human visual system has a quantum efficiency of about 10% with a threshold signal to noise ratio of 4:5. This sensitivity thus is close to the theoretical limit. By comparing the energy of a light quantum with that of a nerve impulse he finds a gain of more than 10^6 . Clearly the light quantum acts as a trigger for a nerve impulse whose energy is provided by the biological system. The usual system thus can be considered as an image converter of highest sensitivity. Yet the materials it uses are completely different from those used in technical image converters.

Another example is provided by the sensitivity of certain fish to electric signals. They make use of such signals in various ways as discussed by Bullok (1977). The lowest electric field that has been observed to evoke a response is of the order of 10^{-8} V/cm. Again in its receptor organ, the fish has none of the materials that would be used for construction of instruments of such high sensitivity.

Controls of activities represent another important set of *in vivo* biological properties. Control of chemical activity is, of course, frequently well understood in terms of relevant enzymes. But these enzymes have to be in their activated state in the right place, at the right time.

Control of cell division, however, is not understood at all. Its absence consists in cancer. We return to this problem in Section IV,D.

Among material properties, most remarkable perhaps is the maintenance of an electric potential difference of about 100 mV across biological membranes. It is well known how this voltage is involved in nerve conduction, but all cells apparently maintain this voltage, leading to the high field of 10^5 V/cm within the membrane (thickness 10^{-6} cm). Such a field leads to strong electric polarization of the materials within the membrane. It is discussed in Section III how this might be involved in the interaction with microwaves.

Of particular interest among materials are enzymes, which through their catalytic action regulate most biological chemical processes. Enzymes are proteins, i.e., they consist of a linear string of amino acids, which in globular proteins are folded in a very definite way. The structure of many enzymes has now been analyzed. They all contain an active group at which,

according to the laws of chemistry, the respective chemical reactions can take place. Yet as shown in a review article by Koshland and Neet (1968), the essential mystery of their enormous catalytic power remains.

Clearly the system as a whole, not only the relatively small active group, should play a role in understanding the action of enzymes. It had been hoped that the spatial arrangement of the amino acids would exhibit a certain spatial order. Nothing of this kind was found, so that according to Monod (1972) knowing the arrangement of 199 amino acids would not permit us to predict to 200th. He concludes that the arrangements are "random."

B. Physics

From the point of view of physics Monod's conclusion is wrong, for physics knows other types of order and organization, besides spatial order. Thus order of motion is widely known and exists in thermal equilibrium (liquid helium II, superconducting electrons) as well as away from equilibrium (lasers, sound waves in air). Another type of order, better termed organization, exists in a well-working machine in contrast to a broken-down one. Thus absence of spatial order need not imply randomness.

Macroscopic organization, to which we alluded in the case of a machine, is, of course, uniquely correlated to details of microscopic structure. But this does not mean that knowing all microscopic details will reveal the interesting macroscopic properties (cf. Fröhlich, 1973a). Not only is the number of microstates so enormous that it cannot be handled but, still more important, the relevant macroscopic properties are expressed in terms of concepts that do not exist in microphysics—they are collective properties.

In the case of an enzyme, after appropriate folding, the system must also act as an enzyme, i.e., it must have very specific properties, although at present they have not yet all been found. F. Fröhlich (1977) has compared the restrictions of the arrangement of words in a language to a restriction of the arrangements of amino acids in an enzyme. Among the known conditions are, in a water-soluble globular protein, the placement of the charged groups on the outside, where they can lower the free energy through their interaction with water. This does not, however, completely determine the folding, which is always exactly the same. If we assume this to be determined through thermodynamic considerations, then the particular conformation must have a free energy sufficiently lower than all the others such that establishment of any of the others becomes extremely unlikely.

This, however, is not the only consideration, because on formation, the linear chain must fold itself, and the initially favorable first steps in folding are not necessarily those that lead to the lowest free energy when folding is completed. We might conjecture that in actually realized enzymes these two types of conditions coincide, so that for them the most favorable pathways also lead to the most favorable free energy. In addition, of course, they will have to have the relevant specific catalytic properties.

One would expect that such conditions select only relatively few linear sequences, but the total number of possible sequences is enormous. Thus since there exist 20 amino acids, the total number of linear arrangements of 200 of these equals $(20)^{200}$. This number, of course, is immense compared with the known number of proteins of 200 amino acids, or even of any length. It is very small, however, compared with the total number of possible microstates in even very small systems. For this reason the use of the concept of information, being equal to negative entropy, in this context cannot be successful. For selection of the one protein (of length 200) would have the information content $200 \log 20$. However, since this does not describe any relevant quality, it is equal in magnitude to small deviations from thermal equilibrium in very small regions. Thus, at temperature T , the imposition of a small temperature difference $\Delta T \ll T$ on a region containing N particles is of the order $N \Delta T/T$.

While negative entropy is not a useful concept for the description of the organization existing in biological and some other systems, one would hope that an appropriate concept does exist, and that it satisfies certain quantitative laws. Such concepts are, of course, used in particular cases, but they have not been unified. Most of them have in common that they involve only relatively few degrees of freedom such, however, that they govern the behavior of the rest.

We presently describe in some detail two physical systems, relevant for biology, in which a certain order is established through the supply of energy (Section III). More frequently, supply of energy to simple physical systems leads to heating, i.e., to disorder, while removal of energy increases order.

An essential feature in our present two cases is the basic importance of nonlinear characteristics in conjunction with a supply of energy. The established new order will be found in the one case to be dynamic in terms of coherent excitation of electric vibrations. This will be seen to lead to long-range selective forces that thus supplement short-range forces, including those leading to standard chemical processes. The selectivity of the long-range forces depends on the frequency of the oscillating systems, which in turn depend on their structure.

A general line of development can thus be seen whereby, e.g., (1) two selected systems attract each other, (2) when meeting they undergo chemical reactions and other rearrangements, (3) this leads to a change of relevant frequencies, and hence (4) to new types of long-range effects.

It is a general feature of active biological systems that energy is always available, through metabolic processes, and that this causes nonlinear changes in molecules, or large subsystems. It is dangerous, therefore, to extrapolate from properties of biological molecules, obtained by extracting them from the living system, to their behavior *in vivo*, although in certain cases this can be done.

The above discussion on the folding of proteins shows that these molecules have remarkable dielectric properties, different in many ways from more ordinary, small molecules (cf. Fröhlich, 1975a). The latter possess a certain vacuum dipole moment that is slightly modified when they are brought into solution. In the case of water-soluble proteins, however, the interaction of these charged groups with water is an essential ingredient in determining the conformation, and hence the dipole moment. Moreover certain groups will become charged, depending on the ion concentration in the water.

This latter also seriously affects the long-range interactions of charged groups. For statically all charges surrounded by cell water are screened within a very short distance through mobile ions. Dipole or higher poles are not necessarily completely screened because short-range interactions may affect positive mobile ions in a different way from negative ones so that the screening in such cases need not be complete. Such screening does not hold, however, in the case of dynamic interaction arising from the fields of oscillating charges in the frequency regions that we consider. Such interaction is governed by the appropriate dielectric constant at the particular frequency. In the range of interest, this dielectric constant depends on frequency but not on wavenumber, whereas the relevant dielectric constant in the static case depends on wavenumber.

Clearly then we must expect particular physical properties of molecules or larger units within a cell to depend on the state and activity of their surroundings, and hence on time. This means that, with some exceptions, we cannot speak of definite molecular properties without referring to the state of the whole cell.

From the point of view of physics we are thus involved in nontrivial nonlinear effects that change with time and are maintained through constant energy supply. As a consequence, nonthermal effects of microwaves should be expected to act at certain frequencies and at certain periods of biological development only. To find these presents a formidable task. Measuring overall properties might thus yield a small effect,

when in fact such an effect might be large but effective during certain periods only.

Thus, from the point of view of physics, biological systems are relatively stable from a microscopic point of view, e.g., the thermal vibrations of single atoms are practically the same as in a corresponding nonbiological system. In some respects, however, involving relatively few degrees of freedom, they are very far from thermal equilibrium, and these degrees of freedom dominate the overall behavior of the rest. They may be described in terms of collective properties or organizations that are carried by a great number of atoms, but use concepts that involve many atoms, i.e., do not exist in individual particle physics. These collective properties evolve as a consequence of supply of energy (metabolism), and usually represent extreme nonlinear displacements. Their detailed characteristics may depend on large-scale configurations and involve activities that in turn will change these configurations, and hence the characteristics of the collective properties.

Among the forces that cause these changes are selective long-range forces and short-range forces, including standard chemical ones. Attention must therefore be drawn to the occurrence of multiple causes. Thus systems might attract each other through selective long-range forces requiring equality of certain frequencies (of Section IV). When they meet, questions of shape may arise to permit certain reactions to start (lock and key principle). Establishment of one does not exclude the other. For this and other reasons, therefore, great care must be taken when transferring methods of procedure in nonbiological sciences to biology.

III. THEORY

In this section we intend to consider on the basis of simple models the two basic collective excitations, relevant to our treatment. They both rest on the high dielectric polarizability of biological materials. One is quasi-static, showing that under very general conditions metastable states with very high electric polarization exist. The second shows that under more stringent conditions coherent electric vibrations may be excited by random metabolic energy (cf. Fröhlich, 1968, 1969, 1977a).

A. Material

Biological membranes have a thickness of about 10^{-6} cm. Oscillations with displacement perpendicular to the surfaces exist so that for the longest wavelength the membrane thickness equals half the wavelength.

Assuming elastic properties corresponding to a sound velocity of 10^5 cm/sec, we find a frequency 0.5×10^{11} Hz. It must be realized now that the membrane usually contains a considerable number of proteins, dissolved in it, so that small sections of the membrane (e.g., between two proteins) might vibrate separately from the rest. In Section II it was mentioned that the membrane is very strongly polarized electrically, in a field of 10^5 V/cm. Hence excitation of vibration of a particular section of the membrane is connected with a vibrating electric dipole. Vibrations with shorter wavelengths, i.e., with higher frequencies, also exist although electrically they would correspond to multipoles. Inhomogeneities within the membrane might, however, yield some dipolar contributions also in these cases.

Individual proteins within the membrane will also be strongly polarized. Their oscillations may also give rise to oscillating dipoles, usually—though not necessarily—with higher frequencies. Optical modes of membranes also contribute to higher-frequency vibrations.

Clearly molecules outside the membranes may, in principle, also oscillate as may other sections of a cell, and a great variety of possible frequencies will exist. Thus Prohofsky and Eyster (1974) by detailed calculation find a great number of giant breathing and rocking modes in double helical RNA, with frequencies reaching down to values below 10^9 Hz. Brill (1978) has estimated a frequency of 1×10^{12} Hz for certain vibrations in blue proteins; Careri (1969) has observed that the stretching vibration of a hydrogen bond lies in a similar frequency region (5×10^{12} Hz). He also observed (Careri, 1973) that a collective excitation might occur in hydrogen-bonded amide structures at about 5×10^{13} Hz. Higher collective modes based on other degrees of freedom may well occur.

Another source of electric vibrations arises from the plasma modes of unattached electrons. Such electrons might find themselves, for instance, in the conduction band of proteins through charge transfer mechanisms (R. Pethig, 1978), who reports a density up to 10^{18} cm $^{-3}$. If m^* is their effective mass and ϵ an appropriate dielectric constant then the plasma frequency ν_p satisfies

$$\nu_p^2 = e^2 n / \pi \epsilon m^* \quad (1)$$

n being their density. Thus a frequency of 10^{11} Hz then requires

$$n \approx 10^{14} (m^*/m) \epsilon \text{ cm}^{-3} \quad (2)$$

It is clear that the actual development of a plasma vibration of this magnitude involves a substantial number of electrons, which at the low density following from (2) would require a volume of the order of a cell's; thus using $\epsilon m^*/m = 10$ would require one electron in a cube of side

length 10^{-5} cm. The dependence of n on ν_p^2 should be remembered here so that for 10^{12} Hz the required volume would be 100 times smaller.

Spectroscopic evidence for frequencies that are integer multiples of about 10^{13} Hz has been obtained by Biscar and Kollias (1973a,b) on specially prepared poly-L-glutamic acid, and may well be a general feature of very high molecular weight compounds.

Now all these possible oscillating systems interact and on the basis of experience from solid-state physics one would expect the appearance of bands of modes. Some of these modes will extend spatially over relatively large regions; others may be restricted to regions around certain molecules—corresponding in solid-state physics to localized vibrations.

Now in contrast to the usual experience in solid-state physics a new feature may arise in our case, due to the large size of the individual systems. As a simplified example (Fröhlich, 1972) consider two large systems at distance R , larger than their linear dimensions L , capable of giant dipole vibrations with circular frequencies ω_1 and ω_2 , respectively. Then through their interaction two frequencies ω_+ , ω_- of the joint system arise,

$$\omega_{\pm}^2 = \frac{1}{2}(\omega_1^2 + \omega_2^2) \pm [\frac{1}{4}(\omega_1^2 - \omega_2^2)^2 + \beta^2/\epsilon_{\pm}^2]^{1/2} \quad (3)$$

where $\epsilon_{\pm}$ is the dielectric constant of the intermediate medium at frequency $\omega_{\pm}$, and β measures their interaction. If, in particular, the two systems contain z_1 and z_2 elastically bound particles of mass M and charge e then

$$\beta^2 = \gamma^2 e^4 z_1 z_2 / M^2 R^6 \quad (4)$$

where γ is of order unity but depends on various angles.

Due to the interaction of the two systems a softened mode ω_- thus arises. If in particular $\omega_1 = \omega_2$, $z_1 = z_2$, and $R = R_0$ such that

$$\omega_1^2 \approx e^2 z / \epsilon_- M R_0^3 \quad (5)$$

then $\omega_-^2 \approx 0$. In this case nonlinear and other neglected interaction terms will have to be employed.

From (3) we note that mode softening will be particularly effective where $\omega_1 = \omega_2$. In this case using $z_1 = z_2 = 100$ and the proton mass for M we have with $R = 10^{-6}$ cm,

$$(e^2 z / M R^3)^{1/2} \approx 10^{13} \text{ Hz} \quad (6)$$

which is indeed very large.

It should be stressed that nonlinear terms arising from the deformation of molecules under the influence of vibrations will lead to further softening when a vibrational mode is strongly excited, as shown later. This

may be of great importance in conjunction with the idea of strong excitation through metabolic activity. Then frequencies occurring in the active (metabolizing) system will differ from those of the resting system. Also the influence of interaction on frequencies (3) imposes a further divergence from the frequencies of extracted molecules.

It should also be remarked that both transverse and longitudinal normal modes exist in an infinite homogeneous medium, and that very few of them are optically active. In our case, however, of strong inhomogeneity together with the existence of inner surfaces, a number of these characteristics are changed. Thus, a giant dipole oscillation of a molecule or of a section of a membrane corresponds to a wavelength of the electric polarization wave, large compared with the linear size of the oscillating region. In this case the division into longitudinal and transverse becomes irrelevant. Due to the existence of surfaces, however, such a system has always some optical activity, even for shorter waves.

B. Metastable Highly Polar States

Consider a finite material, weakly polarized with finite polarization P . An electric self-energy can then be defined (cf. Fröhlich, 1958), which is proportional to P^2 . The proportionality factor depends on the polarizability and on the shape of the material. Deformation of the material will, in general, cause a change in the electric self-energy, which may be positive or negative. A change in elastic energy, due to the deformation, will also occur, but this is always positive if the material was elastically in equilibrium before it was deformed. As a consequence, polarization of a material will cause its deformation, as is well known.

Consider now an example in which the deformation can be described by a single parameter η (Fröhlich, 1969, 1977b). The change U of elastic and of electric energy upon polarization and deformation then has the form

$$U = \frac{1}{2}\gamma P^2(1 + c_1\eta + \frac{1}{2}c_2\eta^2) + \frac{1}{4}C\eta^2 \quad (7)$$

where $C > 0$ and $\gamma > 0$, respectively, are elastic and electric material constants, and the signs of the parameters c_1 and c_2 depend on the particular situation. Minimizing with regard to η for fixed P^2 yields

$$\eta = -c_1\gamma P^2/(c_2\gamma P^2 + C) \quad (8)$$

and hence

$$U = \frac{1}{2}\{\gamma P^2 - \frac{1}{2}[(c_1\gamma P^2)^2/(c_2\gamma P^2 + C)]\} \quad (9)$$

Thus with increasing polarization, U increases more slowly than it would

without deformation. Furthermore, a maximum $U_{\max}$ of U exists provided $c_2 > 0$, and provided

$$2c_2 < c_1^2 \quad (10)$$

at

$$\gamma P_{\max}^2 = C/c_2 \{ [c_1]/(c_1^2 - 2c_2)^{1/2} - 1 \} \quad (11)$$

when

$$U_{\max} = \frac{1}{4}(C/c_2^2)[c_1 - (c_1^2 - 2c_2)^{1/2}] \quad (12)$$

Note that when $|c_2| \ll c_1^2$, then

$$\gamma P_{\max}^2 \simeq C/c_1^2, \quad U_{\max} \simeq \frac{1}{4}(C/c_1^2), \quad |c_2| \ll c_1^2 \quad (13)$$

When $P^2 > P_{\max}^2$ then U will decrease with growing P^2 . At this stage, however, higher powers in P^2 as well as in η are likely to intervene such that a minimum of U is reached at a certain value of P^2 i.e., when the material is strongly polarized. If this minimum $U_{\min}$ of U should be negative then the material would be ferroelectric. If, however, $U_{\min} > 0$, then this highly polarized state is stable against small displacements only and represents a metastable state.

Suppose now that we deal with a large molecule, e.g., a protein, dissolved in a membrane. Assume it to possess a metastable state with dipole moment μ_0 . If this μ_0 has the correct direction, then the system gains the energy $\mu_0 F$ if F is the membrane field. The previously metastable state will then become the stable state if

$$\mu_0 F > U_{\min} \quad (14)$$

It must now be emphasized that displacements corresponding to higher powers in η include what biochemists denote as conformational changes, e.g., large displacements of single ions such as H^+ , or of whole groups. Frequently the energies involved are discussed in terms of localized chemical bonds. This may lead to serious errors as the energy due to relatively small displacements of all the ions may be very substantial. It must be remembered in this context that the interaction between all the electric dipoles arising from the displacement of individual ions does not converge as the total energy arising from this depends on the shape of the molecule.

Another situation arises when the molecule that has been brought into the highly polar metastable state is in solution in the cell water, which contains many small ions. Such ions will then accumulate near the molecule and screen the electric field of the dipole within a very short range.

At the same time this will ensure further stabilization of the metastable state.

As a consequence of this screening, such highly polarized systems will not be involved in long-range interaction. Forces are, however, exerted on ions inside the system, and this may be of considerable importance for reducing activation energies in enzymatic reactions. For systems within membranes, however, this screening is restricted to the regions outside the membrane.

We now quantitatively introduce terms proportional to P^4 in the energy so as to perform the above-mentioned stabilization (Fröhlich, 1973b). At the same time we also consider the dynamic case of giant dipole oscillations. Essentially we thus deal with the case of a space- (x) and time- (t) dependent polarization field $P(t)$, coupled nonlinearly to the longitudinal elastic field $A(x)$. The parameter η must then be replaced by $\text{div } A$, and we use the potential energy density W in such units that if ω_0 is a frequency,

$$W = \frac{1}{2}\omega_0^2 P^2 + (\frac{1}{2}v^2(\text{div } A)^2 + cP^2 \text{div } A)(1 - d^2P^2) \quad (15)$$

Our U in (7) would correspond to the spatial integral over W ; v , c , and d are parameters and the d^2P^2 term has been introduced to ensure convergence at large polarization. Terms corresponding to c_2 terms in U have been neglected. The equations of motion can then be derived in a well-known way from the appropriate Lagrangian density L by (t represents time)

$$\delta \int L d^3x dt = \delta \int (K - W) d^3x dt = 0 \quad (16)$$

where K is the density of kinetic energy of the two fields.

We consider now the case of absence of acoustic phonons so that the kinetic energy of the elastic field can be neglected, and of homogeneous polarization waves so that x dependence of P and $\text{div } A$ does not arise. In this case $\text{div } A$ takes on a time-independent mean value, obtained by differentiating the integrand in (16) with regard to $\text{div } A$. This yields

$$\text{div } A \approx -(c/v^2)p \quad (17)$$

where p is the time average of P^2 ,

$$p = \overline{P^2} \quad (18)$$

and it has been assumed that the fluctuation $(p - \overline{P^2})^2$ is negligible. Equation (16) then yields

$$\ddot{P} + \partial V(P)/\partial P = 0 \quad (19)$$

where the effective potential energy $V(\mathbf{P})$ has the form

$$V(\mathbf{P}) = A + B_1 \mathbf{P}^2 + B_2 \mathbf{P}^4 \quad (20)$$

with

$$A = \frac{1}{2} \frac{c^2}{v^2} p^2, \quad B_1 = \frac{1}{2} \left(\omega_0^2 - \frac{2c^2}{v^2} p - \frac{d^2 c^2 p^2}{v^2} \right), \quad B_2 = \frac{c^2 d^2 p}{v^2} \quad (21)$$

At first sight, (19) with (20) represents a simple anharmonic oscillator. In fact, however, it is highly nonlinear as the potential (20), though (21) and (18), depends on the mean amplitude of the oscillation. As a consequence, for small amplitudes, i.e., small p , the anharmonic B_2 term is negligible and the system oscillates harmonically with frequency ω_0 . Increasing amplitudes introduce some anharmonicity through B_2 but also reduce the frequency by reducing the magnitude of B_1 . Thus simultaneously the mode softens and becomes more and more anharmonic. A drastic change takes place, however, for still larger amplitudes when B_1 becomes negative. The minimum of the effective potential at $P = 0$ now turns into a maximum, but two minima now arise at certain values of P . In this case for a definite amplitude $\pm P_0$, static solutions of (19) exist, i.e., the system is strongly polarized and stable against small displacements. This then represents the metastable state.

At amplitudes at which $p \approx P_0^2$, two branches of vibrations exist. At one the system oscillates weakly around the metastable state; in this case the mean value of P is P_0 . The other case is a large vibration such that the mean value of P disappears, but $\bar{P}^2 \approx P_0^2$.

In the case discussed here, the transition from the nonpolar to the highly polar metastable state thus becomes established when the system is forced into large vibrations through supply of energy. The characteristic feature in our model arises from the behavior of the term B_1 , Eq. (21), which at a certain value of the mean square amplitude p turns from positive to negative values. A similar feature arises in a great number of other processes where supply of energy causes a change of behavior, comparable to a phase transition. Haken (1973) has discussed such questions where the same mathematical description often holds for widely different processes. On the basis of this he suggested a new discipline, synergetics.

C. Excitation of Coherent Vibrations

In this section we come to the crucial consideration that engenders the possibility of trigger action by microwaves. We show that when certain conditions are fulfilled, random energy supply to the modes of a band of electric polarization waves may lead to strong, i.e., coherent, excitation

of a single mode provided the energy supply s exceeds a critical value s_0 (Fröhlich, 1968, 1969, 1977a). If the supply s by metabolic energy is slightly less than s_0 , then a small additional supply, e.g., from microwaves, can trigger off the strong excitation, whose consequences will be discussed in Section IV.

We consider a band of polarization waves consisting of z normal modes with circular frequencies ω_j ,

$$\omega_1 \leq \omega_j \leq \omega_z, \quad j = 1, 2, \dots, z \quad (22)$$

For readers not quite familiar with all aspects of normal modes it must be pointed out that excitation of a vibration with time dependence $\cos(\omega_j t + \varphi)$, say, for a limited time period does not represent excitation of a single normal mode though it can be represented as a superposition of normal modes. Similarly, a vibration that changes its phase φ from time to time does not represent a normal mode because then φ is time dependent. Strong excitation of a normal mode is always coherent.

Our band of normal modes rests on a selected relatively few degrees of freedom in a cell or larger unit. We shall assume it to be embedded in a heat bath of temperature T with which it interacts. As a consequence even when energy is supplied to our modes they tend to be excited according to temperature T as closely as possible. We show that when $s > s_0$ this leads to the strong excitation of a single mode similar to the phenomenon of Bose condensation.

Let n_j be the number of quanta in the mode with frequency ω_j , and let s_j be the rate of energy supply to this mode. Interaction with the heat bath will lead to emission and absorption of quanta by this mode and we consider processes involving one and two quanta. The heat bath is, of course, a very complex system. The interaction will involve dipoles of water and other molecules, mobile ions, certain electronic degrees of freedom, and to some extent also elastic displacements, e.g., when a coupling of the type presented in (15) is involved. The transition probabilities are therefore complicated quantities. The assumption that the heat bath be in thermal equilibrium enforces, however, the principle of detailed balance at its temperature. This means that the kinetic equation for the rate of change of n_j has the form

$$\begin{aligned} \dot{n}_j = & s_j - \phi_j(n_j e^{\beta\omega_j} - (1 + n_j)) \\ & - \sum_l \chi_{jl}[n_j(1 + n_l)e^{\beta\omega_j} - n_l(1 + n_j)e^{\beta\omega_l}] \end{aligned} \quad (23)$$

Both ϕ_j and $\chi_{jl} = \chi_{lj}$ will in general depend on temperature; we use

$$\beta = \hbar/kT \quad (24)$$

We discuss the stationary solutions of this rate equation, i.e., the case $\dot{n}_j = 0$. On summation over j we find for the total rate of supply of energy s ,

$$s = \sum s_j = \sum \phi_j [n_j e^{\beta \omega_j} - (1 + n_j)] \quad (25)$$

On introduction of the excess m_j over the thermal equilibrium distribution n_j^T

$$n_j = n_j^T + m_j, \quad n_j^T = (e^{\beta \omega_j} - 1)^{-1} \quad (26)$$

we obtain

$$s = \sum \phi_j m_j (e^{\beta \omega_j} - 1) < N \phi_{\max} (e^{\beta \omega_{\max}} - 1) \quad (27)$$

where

$$N = \sum m_j \quad (28)$$

and $\phi_{\max}$ is the largest of the ϕ_j . Thus we have a lower limit of the total number N of excess quanta, which increases proportionally to the total rate of supply of energy.

Furthermore, on introduction of μ_j by

$$e^{-\beta \mu_j} = \frac{\phi_j + \sum_l \chi_{jl} (1 + n_l)}{\phi_j + \sum_l \chi_{jl} n_l e^{\beta \omega_l}} \quad (29)$$

we can present a formal stationary solution of the rate equation (23),

$$n_j = \left(1 + \frac{s_j}{\phi_j + \sum_l \chi_{jl} n_l e^{\beta \omega_l}} \right) \frac{1}{\exp[\beta(\omega_j - \mu_j)] - 1} \quad (30)$$

From (29) and (26) we have

$$e^{\beta \mu_j} = 1 + \frac{\sum_l \chi_{jl} m_l (e^{\beta \omega_l} - 1)}{\phi_j + \sum_l \chi_{jl} (1 + n_l)} \geq 1 \quad (31)$$

i.e., $\mu_j = 0$ in thermal equilibrium, $m_l = 0$ (or when $\chi_{jl} = 0$), and hence with (30), since n_j cannot be negative,

$$0 \leq \mu_j < \omega_j \quad (32)$$

We note with (29) that (30) can be rewritten as

$$n_j = \left(1 + \frac{s_j e^{-\beta \mu_j}}{\phi_j + \sum_l \chi_{jl} (1 + n_l)} \right) \frac{1}{\exp[\beta(\omega_j - \mu_j)] - 1} \quad (33)$$

or using (31),

$$n_j = \left(1 + s_j \frac{1 - e^{-\beta \mu_j}}{\sum_l \chi_{jl} m_l (e^{\beta \omega_l} - 1)} \right) \frac{1}{\exp[\beta(\omega_j - \mu_j)] - 1} \quad (34)$$

We now consider the simple case that all ϕ_j are equal to ϕ , and all χ_{jl} are equal to χ . Equation (25) with (26) then becomes

$$s = \phi \sum_j m_j (e^{\beta\omega_j} - 1) \quad (35)$$

From (29) it then follows that all μ_j are equal to μ such that

$$0 \leq \mu < \omega_1 \quad (36)$$

Expression (34) for n_j then reads

$$n_j = \left(1 + s_j \frac{1 - e^{-\beta\mu}}{\chi \sum_l m_l (e^{\beta\omega_l} - 1)} \right) \frac{1}{\exp[\beta(\omega_j - \mu)] - 1} \quad (37)$$

or making use of (35),

$$n_j = \left(1 + \frac{s_j}{s} \frac{\phi}{\chi} (1 - e^{-\beta\mu}) \right) \frac{1}{\exp[\beta(\omega_j - \mu)] - 1} \quad (38)$$

Thus using (26), (28), and

$$N^T = \sum n_j^T \quad (39)$$

we find

$$\begin{aligned} N + N^T &= \sum_j \left[1 + \frac{s_j}{s} \frac{\phi}{\chi} (1 - e^{-\beta\mu}) \right] \frac{1}{\exp[\beta(\omega_j - \mu)] - 1} \\ &\leq \left[1 + \frac{\phi}{\chi} (1 - e^{-\beta\omega_1}) \right] \sum_j \frac{1}{\exp[\beta(\omega_j - \mu)] - 1} \end{aligned} \quad (40)$$

because $s_j \leq s$ and $\mu < \omega_1$.

If in particular the supply s_j to all modes is the same, then $zs_j = s$, and

$$N + N^T = \left[1 + \frac{\phi}{z\chi} (1 - e^{-\beta\mu}) \right] \sum_j \frac{1}{\exp[\beta(\omega_j - \mu)] - 1} \quad (41)$$

Dependence of both (40) and (41) on s is implicit through μ only. Furthermore, through (27), N must increase with growing supply s . In both (40) and (41), once N , i.e., s , has surpassed a critical value N_0 , s_0 , this is possible only if μ closely approaches ω_1 [cf. (36)] in which case n_1 will become very large.

This is exactly the situation met in Einstein condensation of a Bose gas that arises in equilibrium below a certain critical temperature. In our case the corresponding phase transition is enforced by the supply of energy—again a specific case of Haken's (1973) rules in synergetics.

It will be seen from (29) that $\mu_j = 0$ when the $\chi_{jl} = 0$. It is thus the two-quanta interaction with the heat bath that is mainly responsible for the condensation; large χ favors our effect.

In the more general case we make use of (27) and write (34) in the form

$$n_j = \left(1 + \frac{s_j}{s} (1 - e^{-\beta\mu_j}) \frac{\sum_l \phi_l m_l (e^{\beta\omega_l} - 1)}{\sum_l \chi_{jl} m_l (e^{\beta\omega_l} - 1)} \right) \frac{1}{\exp[\beta(\omega_j - \mu_j)] - 1} \quad (42)$$

instead of (38). Here the factor of s_j/s in general varies only slowly with s . Forming $\sum n_j$, the same argument as that following (41) holds again. In the present case, however, only one of the μ_j need approach ω_j very closely, in which case the particular n_j becomes very large. From (31) we would conclude that if for a particular j the transition probability ϕ_j is much smaller than that of all other states then its μ_j may approach ω_j before the others and therefore this particular n_j will be much larger than the others. On the other hand, supplying energy to a particular state only does not necessarily mean that this state will be strongly excited, for the χ_{jl} transitions distribute energy appropriately over all states and, as remarked before, large χ favors our effect.

It therefore seems reasonable to neglect the s_j term in the expression for n_j . Going back to (38) then, we have approximately

$$n_j \approx \{\exp[\beta(\omega_j - \mu)] - 1\}^{-1} \quad (43)$$

and through (35),

$$z + s/\phi = \sum_j (e^{\beta\omega_j} - 1) / \{\exp[\beta(\omega_j - \mu)] - 1\} \quad (44)$$

Clearly $s = 0$ implies $\mu = 0$, while for sufficiently large s/ϕ we can neglect z . μ is then determined by s/ϕ . In the usual way, transforming the sum into an integral, assumed to converge as μ approaches ω_1 , one finds s_0 for $\mu = \omega_1$. Condensation into ω_1 then arises for $s > s_0$. In this approximation it is irrelevant into which states the energy is fed as the large value of χ leads to immediate distribution.

If now through metabolic processes the value of s is below s_0 , say

$$s = s_0 - \Delta s \quad (45)$$

then supply of energy at the small rate Δs into any of the modes ω_j is sufficient to cause condensation. Also if $s > s_0$, then energy supplied into any of the ω_j will be largely channeled into the condensed state.

Rate equation (23) can, of course, be derived from more explicit models, as has been shown by Bhaumik *et al.* (1976) and by Wu and Austin (1977, 1978a) using different interaction terms between the electric vibrations and the heat bath (cf. also Mills, 1979). Both impose temperature equilibrium on the heat bath, so that detailed balance at this temperature, and hence Eq. (23), must arise. These papers permit, however,

giving explicit interpretation to the transition probabilities ϕ and χ , although so far they have not dealt with the important cases of water dipoles and of unattached ions in the heat bath.

Wu *et al.*'s interaction term is closely related to our $P^2 \text{ div } \mathbf{A}$ term in Eq. (15) if the elastic $\mathbf{A}$ field represents the heat bath. Such a term, in view of P^2 , leads to two-quantum processes, not only to a combination of absorption and emission, as in the χ processes, but also to absorption or emission of two quanta. Probability for such terms, however, is much smaller than of χ processes as Wu and Austin (1978b) prove in detail. This arises because in the χ processes the energy $\hbar(\omega_j - \omega_l)$, in the other processes, however, the energy $\hbar(\omega_j + \omega_l)$, has to be exchanged with the heat bath, if quanta with frequencies ω_j and ω_l are involved. In addition, the total wave vector must be conserved, which in view of the small magnitude of the wave vector of our polar modes permits only very small energy transfer. This is always possible for the difference $\omega_j - \omega_l$ but not for the sum. The reader is reminded at this stage that in analogy to the case of ionic crystals, a very low wave vector of polarization waves (very long wavelength) implies a very high wave vector of displacements of mass, as positive and negative ions then move in the opposite direction.

It is easy to see that a similar result holds for the interaction of polarization waves with unattached ions, say of mass M , energy kT , and wave vector $\mathbf{k}$. The important point here is that $\mathbf{k}$ is very large, of order 10^8 – 10^9 cm^{-1} if M is the proton mass. The transferred wave vector $\mathbf{g}$ has very much smaller magnitude. Thus the wave vector g of a giant dipole oscillation is $g = 0$; for a mode carried by interacting oscillating systems $1/g$ equals at least their mean distance. Thus through the interaction, $\mathbf{k}$ changes into $\mathbf{k} + \mathbf{g}$, so that $(\mathbf{k} + \mathbf{g})^2 - k^2 = g^2 + 2kg \cos \theta$ where θ is the angle between $\mathbf{k}$ and $\mathbf{g}$. Conservation of energy then yields

$$(\hbar^2/2M)(g^2 + 2kg \cos \theta) = \hbar(\omega_j \pm \omega_l) \quad (46)$$

and hence since the g^2 term can be neglected

$$|\cos \theta| = \frac{1}{2} \frac{\hbar|\omega_j \pm \omega_l|}{kT} \frac{k}{g} \leq 1 \quad (47)$$

Clearly this can always be satisfied for $\omega_j - \omega_l$, i.e., a combination of absorption and emission, in spite of the very large magnitude of k/g . The $\omega_j + \omega_l$ case, however, is largely excluded unless frequencies very much lower than 10^{11} Hz were contemplated.* This also holds for ϕ -type single-quantum emission processes.

* This discussion also shows that Livshits' (1977) dogmatic conclusion is incorrect.

It should be emphasized at this point that the required importance of χ transitions represents a material property that we would expect to have been established in the course of evolution of biological materials. It is easy to demonstrate possibilities that counteract Bose condensation. As a simple example we might assume the absence of χ processes, but the presence of strong interaction between polarization waves without the help of the heat bath with which they are assumed to be weakly coupled through ϕ processes only. In this case, the supply of energy would simply result in hot polarization waves without strong excitation of a single mode. Thus we have shown that depending on certain microscopic properties, the supply of energy either may make a system hot or may result in the creation of a new type of order—a feature very important in many other aspects of biology.

Theoretical investigation of the quantitative detailed microscopic requirements would be extremely difficult. At the present time, therefore, it should be left to experiments to decide on the existence of coherent excitations. Present evidence is strongly in favor. It was the task of the theory to point out the possibility and to pursue its consequences.

It should also be pointed out that time-dependent solutions of the kinetic equation (23) will be of considerable importance for estimates of the time required to establish the strongly excited state. A first step in this direction has been taken by Wu and Austin (1979).

IV. CONSEQUENCES

A. General

The theoretical models discussed in Section III are based on the dielectric behavior of biological systems in conjunction with nonlinear excitation. One of them shows that single modes of electric vibrations can be strongly excited provided certain conditions are fulfilled. The other illustrates under rather more general conditions, that strongly polarizable systems will possess metastable states with very high polarization. It has also been shown that interaction between two systems may lead to mode softening. Clearly if these possibilities are realized in biological systems, then system properties of a much more complex nature may be considered. Among these, the establishment of the strongly excited mode as a limit cycle might be contemplated; this would be relevant in overcoming the friction of the surroundings. A further possibility to be contemplated is the excitation of solitons. It may be remembered that some nonlinear

vibrating systems possess very stable dynamic excitations (solitons or solitary waves). Such excitations provide the possibility of nearly frictionless transport of energy. Theoretical work on this possibility is in progress based on Eq. (15) (Bilz *et al.*, 1980).

In connection with our vibrational model it should first be remembered that the expected frequencies are $10^{10} - 10^{11}$ Hz for membranes, $10^{12} - 10^{14}$ Hz for proteins or more general for certain bond-stretching groups, 10^9 Hz for DNA or RNA molecules. Mode softening due to interactions may, however, open the range to much lower frequencies. Also frequencies of the order of 10^{15} Hz arising from oscillations of bound electrons cannot be excluded in principle.

It must be mentioned in this context that stored energy is available in active biological systems in the form of "high-energy" chemical bonds, and electrically in the strong fields in membranes. To this we wish to add, tentatively, the energy of vibrations of polar modes, maintained during certain periods through energy supply, at a rate s , from various sources of metabolic energy. Suppose this rate s to be slightly smaller than s_0 , then minimum energy supply required for strong excitation of a single mode [cf. (45)], so that the system is prepared for appropriate action by further supply of energy at a very low rate ($> s_0 - s$). If the band of polar modes has frequencies in the microwave region, then external supply of energy by irradiation with microwaves will effect the coherent excitation and hence prematurely start the biological processes connected with it. From Section III,C it follows that the frequency of the actually excited coherent mode may differ from that of the incident microwave—it may be lower or higher—and microwaves with quite a number of different frequencies may excite the same coherent mode and thus influence the same biological process.

Actual experimental specification of the particular biological process will require application of the microwaves at specific instances of development of the biological system; whether in an overall measurement—say of the rate of growth—this leads to an increase or a decrease will depend on the particular type of event triggered by the microwaves.

While experimental steps required for this specification may involve considerable difficulties, investigation of the dependence on intensity and on frequency is imperative and should be possible with the use of straightforward techniques.

The main energy required for the various excitations must come from metabolic processes—biological pumping. Such processes are frequently connected with chemical activities such as protein synthesis. This invariably leads to displacement of ions, hence to electric signals seen by all polarizable regions, and hence to the possibility of energy exchange.

B. Long-Range Interaction

Static electric charges are screened by the ions of cell water at very short distances. Excitation of the highly polar state in a system, e.g., a protein, will therefore not lead to long-range interaction with another one when the systems are in cell water, although it will lead to effective interaction when the systems are placed within a membrane. In this latter case, due to the membrane field, the dipoles of the two systems are expected to be parallel in the direction of the field. The interaction is then repulsive.

Formally, the interaction between such systems is governed by the dielectric constant of the medium, in the case of static charges in cell water by the appropriate longitudinal dielectric constant. As a consequence, the potential arising from a static point charge becomes a screened Coulomb potential.

This is quite different for charges oscillating at sufficiently high frequencies when screening becomes only effective at distances considerably larger than a cell diameter. Dielectric loss must, however, also be taken into account, which it will be remembered is large in water in the 10^{10} Hz region.

Consider now the interaction of two systems at a distance R , sufficiently larger than their linear dimension L , capable of polar vibrations (Fröhlich, 1972). On this condition interaction between these giant dipole vibrations only has to be considered as the fields due to higher poles decrease rapidly with distance. The band of polar modes of the combined system thus consists of the dipole, as well as of the higher pole vibrations. To calculate the interaction, at distance R , however, it is sufficient to select the modes with frequencies ω_+ and ω_- given in (3). This actually is a simplification assuming both systems to oscillate along one direction only, but this will not alter the basic results.

Assume now the mode with frequency ω_- to be strongly excited to energy E_- . If the vibration is (nearly) harmonic then this energy is proportion to ω_-^2 and to the square of the amplitude. We may consider this latter to remain unchanged on a virtual displacement of R . The force $\mathbf{f}$ between the two systems is then given by

$$\mathbf{f} = - \frac{\partial \omega_-^2}{\partial \mathbf{R}} \frac{E_-}{\omega_-^2} \quad (48)$$

where from (3)

$$2\omega_-^2/(\omega_1^2 + \omega_2^2) = (A^2 + R_0^8/\epsilon R^6)^{1/2} \quad (49)$$

The constant distance R_0 follows at once from (4), and we use, assuming $\omega_1 > \omega_2$,

$$A = (\omega_1^2 - \omega_2^2)/(\omega_1^2 + \omega_2^2) > 0 \quad (50)$$

Consider now the case that A^2 is large compared with the following term, so that

$$2\omega_-^2/(\omega_1^2 + \omega_2^2) \approx 1 - A - R_0^3/2\epsilon_-R^6 \quad (51)$$

In the case of resonance, on the other hand, when $\omega_1 = \omega_2$, i.e., $A = 0$,

$$\omega_-^2/\omega_1^2 = 1 - R_0^3/\epsilon_-R^3 \quad (52)$$

and the interaction becomes, assuming $R_0/R > 1$,

$$I = -(R_0^3/\epsilon_-R^3)E_- \quad (53)$$

Owing to the pumping, E_- is expected to be large compared with kT , and the interaction can therefore be considerable in the resonance case. It is of long range ($1/R^3$) and attractive if $\epsilon_- > 0$. In the absence of resonance (51), the interaction decreases rapidly with distance as $1/R^6$.

If we consider the case that mode ω_+ rather than ω_- is excited, then repulsion arises provided $\epsilon_+ > 0$. Resonance interaction, $\omega_1 = \omega_2$, is again of long range $\propto 1/R^3$.

Equation (3) for $\omega_\pm^2$ is no longer valid when R is less than the linear dimension L of the subsystems. A further condition on the validity of the above is $\omega_\pm^2 > 0$, which also imposes a limit on R .

When R is much less than L , then roughly speaking the term $1/R^3$ in β will be replaced by the smaller $1/L^2R$. In addition, however, nonlinear terms may be involved as discussed by Bhaumik *et al.* (1978).

In pursuing further the quantitative properties of the interaction, it must be realized that the dielectric constant depends on frequency. Equation (3), therefore, represents an implicit equation for $\omega_\pm$ as has been realized by Genzel (1978), who solved it by using the appropriate expression for the dielectric constant of water.

It must also be pointed out here that in the resonance case, $\omega_1 = \omega_2$, the large splitting $\omega_+ - \omega_-$ may imply substantial differences between ϵ_+ and ϵ_- , and as a consequence lead to abnormally large interaction even if none of the modes is excited beyond thermal equilibrium.

In this case, the interaction energy I is defined as the difference of the free energy at distance R from its value at $R \rightarrow \infty$. This yields

$$I = \begin{cases} [(\omega_+ + \omega_-) - (\omega_1 + \omega_2)]\hbar/2, & \text{if } \hbar\omega \gg kT \\ kT \log \omega_+\omega_-/\omega_1\omega_2, & \text{if } \hbar\omega \ll kT \end{cases} \quad (54)$$

$$(55)$$

Using the resonance case (52), and similarly with ω_+ , we find

$$\omega_{\pm} = \omega_1(1 \pm R_0^3/2\epsilon_{\pm}R^3), \quad \text{if } R_0^3 \ll \epsilon_{\pm}R^3 \quad (56)$$

and hence,

$$I = \frac{\hbar\omega_1}{4} \frac{R_0^3}{R^3} \left(\frac{1}{\epsilon_+} - \frac{1}{\epsilon_-} \right), \quad \text{if } \hbar\omega \gg kT \quad (57)$$

In the case of (55), the factor $\hbar\omega_1$ in (57) must be replaced by $2kT$. Thus when $\epsilon_+ \neq \epsilon_-$, a long-range $1/R^3$ interaction arises even in thermal equilibrium. The case $\epsilon_+ = \epsilon_-$, when the $1/R^3$ interaction vanishes, leads to the well-known $1/R^6$ London interaction.

The long-range interaction following from (53) may have considerable biological significance. Being frequency selective, it can lead to long-range recognition, and attraction if ω_- is excited. This might, for instance, hold for the attraction of an enzyme to a particular site, when the appropriate frequency on that site, interacting selectively with the particular enzyme, has been excited.

Holland (1972) has applied this idea to the pairing of homologous chromosomes in meiosis. This must arise from a selective long-range attraction, which according to Watson (1970) has until now remained "a great mystery." He also suggests that this interaction (he denotes it as dynamic specificity) might help to understand how RNA molecules, and proteins after their synthesis, get to their destinations in the cell.

Excitation of ω_+ , as we have seen, leads to selective long-range repulsion. We might conjecture this to be of relevance for the recognition of parts of a biological entity by the immune system. Each biological entity might have its own characteristic frequency or frequencies, which its immune system recognizes by being repelled from it.

Long-range selective attraction will lead, of course, to close approach of the systems involved. At this stage, we have seen, mode softening of ω_- will take place. As a first consequence the combined system changes its long-range characteristics; the resultant oscillation may or may not be dipolar, depending on the angle between the oscillating dipoles of the two subsystems. Furthermore, the interaction with other representatives of the two subsystems will no longer be of the long-range type (53) because of the change of frequency. After the close approach of the two subsystems, short-range chemical processes of the usual type will be initiated. Thus, possibly such systems may exhibit both short- and long-range selectivity.

The mode softening arising from the approach of an enzyme to its substrate will be counteracted to some extent by nonlinear terms that have been neglected in the calculation of ω_- in Eqs. (3) and (4). One possibility

of achieving this would be in terms of the model discussed in Section III,B although this has not been carried out in detail. As a consequence we may expect a state with higher frequency and with strong polarization to arise, stabilized by mobile counterions that screen the arising dipole moment. It has been conjectured (cf. Fröhlich, 1970) that the active states of enzymes always possess this high polarization, leading to strong internal electric fields. Independently, it has been pointed out by Green (1974) that these fields will be capable of reducing activation energies, and hence help understanding the enormous catalytic power of enzymes.

It should also be emphasized again that the strong mode softening that arises at a relatively large distance R_0 [cf. (5)] is a characteristic of large molecules possessing giant dipole vibrations, as we encounter them in biology. Thus a frequency of 10^{12} Hz with $z = 30$, using the proton mass, yields $R_0 \approx 10^{-6}$ cm.

A further consequence of the nonlinear terms in (15) arises from the possibility of the excitation of solitons (solitary waves) in biological systems such as enzymes (Bilz *et al.*, 1980). Solitons are localized excitations with great stability and thus are capable of transmitting energy nearly loss free when no chemical is transported. This may arise, e.g., in the action of muscles.

Finally it should be pointed out here that the dielectric function of the medium between two interacting systems—as it arises in various expressions such as (57)—is the longitudinal one for the appropriate wave-numbers and frequencies $\epsilon(k, \omega)$. This may differ considerably from the transverse dielectric constant measured at this frequency.

C. Models for Nonthermal Actions of Microwaves

A microwave penetrating into a medium loses energy to it, which in the realm of linear response is proportional to the intensity of the radiation and whose magnitude and frequency dependence is governed by the complex dielectric constant of the medium. It must be understood (e.g., Fröhlich, 1958) that in a medium like water the energy loss arises through a subtle influence of the field of the microwave on the thermal motion of the electric dipoles of water. As a consequence this thermal motion is only very minutely perturbed, but an increase in temperature must arise so that under stationary conditions as much energy is removed by heat conduction as is supplied by the absorption of the microwave. Consequently we should not necessarily expect the occurrence of nonthermal effects in regions of high dielectric absorption.

Effects are known in physics in which the supply of external energy is effected to a particular subsystem that is coupled weakly to the rest of the

system. This subsystem then may still be near a thermal equilibrium corresponding to a temperature higher than that of the rest. Hot electrons in semiconductors is a well-known example, whereby the electrons receive energy from an external electric field. Their density must be sufficiently high to distribute through mutual collisions the energy they receive among themselves before passing it on to the lattice vibrations, which are treated as a heat bath at fixed temperature.

In our model of coherent excitations (Section III,C) external energy is supplied to the system of polar vibrations. They interact with a heat bath in two ways, either by exchanging quanta $\hbar\omega_j$ (ϕ processes) or by a kind of phonon Raman effect by exchanging only the difference $\hbar(\omega_i - \omega_j)$ of the energy of two quanta (χ processes). This latter effect is considered to be more prominent than the first. The constituents of the heat bath that interact strongly with the polar modes are unattached ions, mobile dipoles, and to some extent elastic vibrations. The interaction with unattached ions has a well-investigated analog in the interaction of nearly free electrons with the polar modes of an ionic crystal. Here emission of single quanta $\hbar\omega$ by a free electron is a rare event due to the difficulty of conserving energy and momentum, so that the principal scattering exists in double processes (emission + absorption). The same holds in our case as pointed out with respect to ϕ processes in connection with (47).

In addition to these processes, it has been assumed that processes through direct interaction of two polar vibrations are negligible compared with χ processes. Under these circumstances it has been shown that χ processes impose a distribution of excitation in the band of polar modes that is governed by the temperature T of the heat bath. In the approximation in which we consider supply of energy s and loss of energy ϕ as slow processes, this implies that

$$n_j(1 + n_i)e^{\beta\omega_j} = n_i(1 + n_j)e^{\beta\omega_i} \quad (58)$$

or

$$n_j e^{\beta\omega_j} / (1 + n_j) = n_i / (1 + n_i) e^{\beta\omega_i} \quad (59)$$

must hold, i.e., that in zero order the χ terms of (23) must vanish. From this follows at once our approximative solution (43), and hence strong excitation of a single mode when the rate of supply of energy s exceeds a critical one s_0 . In this approximation there exists a single "chemical potential" $kT\mu$ and ω_1 is the excited mode. The more refined model yielding a different μ_j for each ω_j [cf. (42)] does not permit the crude approximations made above.

We now consider the quantities that determine the magnitude of s_0 as follows from (35) or, in the simple approximation (43), from (44). Clearly

μ must increase as s grows, starting with $\mu = 0$ for $s = 0$. Strong excitation of ω_1 arises when

$$\omega_1 - \mu < \omega_2 - \omega_1 \quad (60)$$

How sharp a transition in n_1 as a function of s at a certain s_0 will be depends on the distribution of ω_j levels and cannot be answered in general. Such details are not very relevant for our present purposes. It is most important, however, that Eq. (44) does not yield s_0 , but rather the ratio s_0/ϕ . Thus the smaller ϕ , the lower the energy transfer from the polar modes to the heat bath, the smaller is the critical rate s_0 of energy supply at which the transition to strong excitation arises. Thus when $s > s_0$, then our system is capable of storing energy until the strong excitation of the mode ω_1 , containing under stationary conditions the energy $n\hbar\omega_1$, is achieved. In principle, thus, very low ϕ will permit very small $s > s_0$ to achieve strong excitation. Such storage is also characteristic for signal over noise systems.

We consider now a biological process that is initiated by the consequences of strong excitation of a certain mode, leading to selective long-range attraction. This might, for instance, lead to an enzyme being attracted to an appropriate position in which it becomes active, or to another of the possibilities suggested above. Let t_1 be the time required for this process, which then is assumed to be followed by a chain of biological processes taking a time t_2 . The rate B of the total is thus

$$B = (t_1 + t_2)^{-1} \quad (61)$$

If t_2 does not involve long-range interaction, then B will depend on it through t_1 only. The first process then requires excitation of the coherent mode through energy supply satisfying $s > s_0$. The larger $s - s_0$, the shorter will be t_1 , and we approximate this by

$$t_1 = a/(s - s_0)^n \quad (62)$$

where a is a constant and $n \geq 1$ a positive number. It follows that

$$B = \frac{1}{t_2 (a/t_2) + (s - s_0)^n} \quad (63)$$

If $s < s_0$ then long-range attraction is absent so that the required transport can occur through diffusion only, making t_1 very large and B very small. Increased energy supply, on the other hand, will make t_1 very small so that, ultimately, the process described by t_2 determines the rate B , which becomes independent of s . Thus the function $B(s)$ is steplike with the step at $(s - s_0)^n = a/t_2$; it is zero if $s < s_0$.

Consider now s to be composed of a biological part s_B and of s_m , the rate at which energy is supplied by microwave irradiation,

$$s = s_B + s_m \quad (64)$$

Then if $s_B < s_0$, we find $B = 0$ for small s_m until $s_m = s_0 - s_B$, when B rises with increasing s_m , at first very slowly, and when s_m reaches the value at which $(s_m + s_B - s_0)^n = a/t_2$ in a steplike manner, after which B no longer changes with increasing s_m . The rate of the biological process described by B is thus either unchanged by microwave irradiation or increased, but never decreased. It must be pointed out, however, that this does not imply that biological processes of a kind different from that involved in B are not influenced. In fact, if the microwave initiates a process that at the particular time is not appropriate to the action of the biological system, then it may be expected to disturb the general development and in an overall observation, such as rate of growth, have a negative influence.

Clearly it is desirable then to test the influence of microwave radiation on a specific biological property, described in terms of B . It might be hoped, however, that even overall measurements will indicate the dependence of $B(s)$ on s_m .

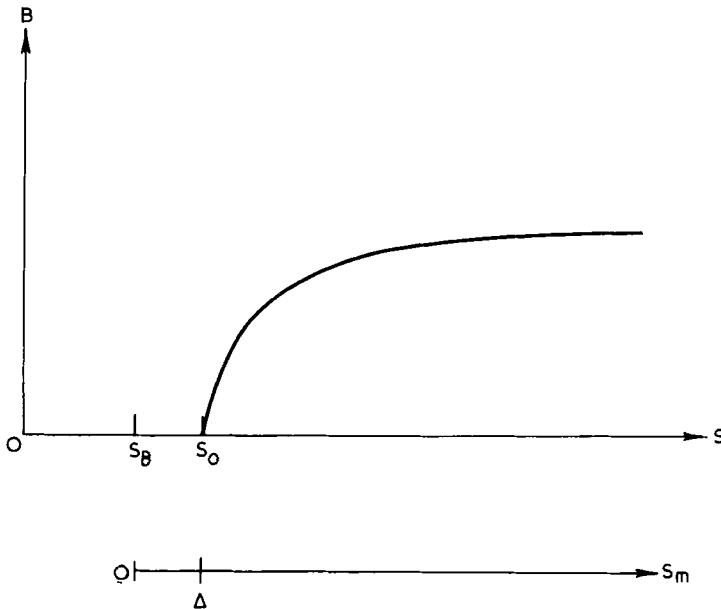


FIG. 1. Theoretical dependence of the rate of response B on the rate of energy supply S . The dependence of B on the rate of supply of radiation energy s_m is indicated if the biological energy supply is at the rate s_B .

Figure 1 shows $B(s)$ as a function of s . For given s_B , this also represents the dependence of B on the intensity of radiation s_m , if $s_m = 0$ is placed at $s = s_B$. Hence if $s_B > s_0 + a/t_2$, then the action of external radiation s_m will be negligible. If, however, s_B is less than s_0 , then the action of the radiation again follows a steplike function of s_m ; it acts as a typical trigger effect if s_B falls just below s_0 . The actual value of s_m , at which the biological effect is initiated may be very small, even if $s_B = 0$, provided ϕ is sufficiently small as discussed above.

It should be pointed out, at this stage, that the nonlinear response to s_m implies that pulsed radiation may yield a response different from continuous radiation of the same mean intensity. Thus, e.g., if the mean intensity is such that $s_B + s_m < s_0$ then continuous radiation will have zero response. The maxima of pulsed radiation may, however, reach above s_0 and thus yield a response.

It has been pointed out above that when $s > s_0$, radiation absorbed by any of the modes of the polar vibrational band will transfer its energy to the strongly excited mode. If this mode represents giant dipole vibrations, then one might hope to be able to detect it by absorption and reflection measurements. It must be realized, however, that these properties are largely dominated by cell water. The biological subsystems involved in a particular type of coherent vibrations might use a very small volume only, and thus contribute only little to the overall optical properties. It is difficult, at present, to predict whether such measurements could have a chance of success.

D. Multicomponent Systems and the Cancer Problem

Consider an assembly of equal units capable of polar oscillations. In crystal dynamics it is shown that this gives rise to a band of polar normal modes (polarization waves) whose lowest frequency is different from zero. These modes can be divided into longitudinal and transverse modes; furthermore, three modes with zero wavenumber, i.e., giant dipole vibrations, exist whose frequencies depend on the macroscopic shape of the material (cf. Section 18 of Fröhlich, 1958). This arises from the long range of the interaction between polar vibrations, as has also been discussed in Section III,A for the case of two units.

A noncrystalline arrangement of the units does not basically change these conclusions although it influences the dependence of frequency on wavenumber. In general if all units have equal frequency ω , then the band of normal modes ranges from frequencies lower than ω to frequencies higher than ω , just as in Section III,A ω_- is lower and ω_+ higher than the original frequency. It is of interest in this context that in ω_- , the two

systems oscillate in phase when the line connecting the two systems is parallel to their direction of oscillation, while in the case that it is perpendicular they oscillate with opposite phase. In the first case then ω_- represents a giant dipole vibration, while in the second it does not. We can conclude that in large disordered systems no general prediction of the position of giant dipole vibrations within the band can be made.

We now apply the considerations of Section III,C to our band. To start with, we discuss the simple case in which the rate of supply of energy s is small, $s < s_0$, and neglect the nonlinear χ terms in (23). We ask for the behavior of a single of our units, whose frequency in isolation we denote as ω . This unit interacts with the (weak) excitations of various normal modes, frequencies ω_j , in a linear way so that a linear superposition of vibrations at various frequencies ω_j is excited. For a single normal mode then, the amplitude x of the oscillation of our single unit will be proportional to the amplitude A_j of the normal mode so that

$$x = \alpha_j A_j, \quad A_j = A_{j0} \cos \omega_j t \quad (65)$$

where A_{j0} is time independent and the parameter α_j depends on the geometry of our system and on the wavenumber of the normal mode. Also then

$$\ddot{x} + \omega^2 x = \beta_j A_j \quad (66)$$

where the parameter β_j also depends on the geometry of our system and on the wavenumber of the normal mode. The right-hand side is proportional to the force exerted by all the units on our particular one when the normal mode j is excited. From (65) thus

$$x(\omega^2 - \omega_j^2) = \beta_j A_j \quad (67)$$

and using (65) we find the frequency

$$\omega_j^2 = \omega^2 - \beta_j / \alpha_j \quad (68)$$

In the case of crystals, the parameters β_j and α_j can easily be calculated and yield ω_j as a function of the wavenumber. At weak excitation, the oscillations of the various normal modes superimpose linearly, and correspondingly x becomes a linear superposition of the terms presented in (65). The realistic case of interaction with a heat bath implies introduction of a damping term and stochastic supply of thermal energy and results in a Planck distribution. Further supply of external energy to various normal modes leads to additional excitations, and if this supply does not exhibit any phase correlation, for the various frequencies, then x will be a linear superposition of contributions at the frequencies ω_j of all the modes, showing no phase correlations.

Consider now the opposite extreme, the case that the energy supply is large, $s > s_0$ and that the nonlinear χ terms are important. As a result, a particular mode with frequency ω_b , say, undergoes Bose condensation so that it is very strongly excited to the amplitude A_b , which we may assume to be so large that the temporal mean value of A_b^2 is large compared with the sum over the A_j^2 of all the other modes, so that we can disregard their excitation. As a consequence, the amplitude x of our particular unit will also be strongly excited, and we wish to investigate the stability of this excitation. For this purpose, introduction of the nonlinear χ terms into the dynamic equations of the system would be required, a task that has not yet been carried out. We note, however, that when the χ_{jl} are very large, then they nearly entirely regulate the magnitude of the strongly excited mode; in other words, direct supply into and loss of energy from this mode through s_b and ϕ_b are of minor importance. It should therefore be a good approximation to supplement Eq. (66) with nonlinear terms that enforce the required excitation,

$$x = \alpha_b A_b \quad (69)$$

for given A_b . This can be done by extending (66) into the limit cycle equation

$$\ddot{x} + \omega^2 x + \gamma \dot{x}(x^2 - \alpha^2 A_0^2/4) = \beta A_0 \cos \omega_b t \quad (70)$$

where γ is a positive constant, and, for simplicity, we have omitted a subscript b in α , β , and A_0 . Solution of this well-known equation is, of course, no longer strictly harmonic. However, we deal with the first harmonic only, and disregard higher frequency terms. Clearly the γ terms lead to damping when x^2 is large and to antidamping when this term is small, thus driving the system into a definite amplitude determined by A_0 . In fact under stationary conditions when [cf. (65)]

$$x = x_0 \cos \omega_b t, \quad x_0 = \alpha A_0 \quad (71)$$

we find

$$\begin{aligned} \dot{x}(x^2 - \alpha^2 A_0^2/4) &= -\omega_b x_0 \sin \omega_b t (x_0^2 \cos \omega_b t - \alpha^2 A_0^2/4) \\ &= -\frac{1}{4} \omega_b x_0 (x_0^2 - \alpha^2 A_0^2) \sin \omega_b t = 0 \end{aligned} \quad (72)$$

if we disregard $\sin 3\omega_b t$ terms. Thus at the amplitude given by (71), and at this amplitude only, Eq. (70) reduces to (66) with $j = b$.

To investigate now the stability of our solution we consider x to deviate from it by ξ , i.e., replace (71) by

$$x = \alpha A_0 \cos \omega_b t + \xi \quad (73)$$

Neglecting terms in ξ^2 , Eq. (72) is now replaced by

$$\begin{aligned} & [\alpha A_0(-\omega_b \sin \omega_b t) + \dot{\xi}][(\alpha A_0)^2 \cos^2 \omega_b t - (\alpha A_0)^2/4 + 2\alpha A_0 \xi \cos \omega_b t] \\ & = \frac{1}{4}(\alpha A_0)^2 \ddot{\xi} + \dots \end{aligned} \quad (74)$$

if we neglect terms with frequency $2\omega_b$. As a consequence we find from (70)

$$\ddot{\xi} + \omega^2 \xi + \frac{1}{4}\gamma(\alpha A_0)^2 \dot{\xi} = 0 \quad (75)$$

the equation of a damped oscillator whose effective damping constant is larger the higher the excitation of the Bose condensed mode. Thus any deviation ξ of the excitation of our unit from that required by the mode quickly dies out, except for higher frequency terms, which we neglect here. Similarly, additional energy has to be spent in order to cause a deviation ξ from the stationary x .

It should be pointed out now that excitation of a normal mode ω_j also induces vibrations in systems whose frequencies in isolation differ from ω ; in fact, the normal mode embraces all of these. The excitation of single regions of such systems will, however, be much smaller than those of our units unless their frequencies in isolation come very close to ω . In general, therefore, the above considerations do not apply to such different regions. A more quantitative statement must, however, wait for a more detailed treatment.

Two types of consequences arise from the strong excitation of a vibration of frequency ω_b in an assembly of (nearly) equal units, e.g., a tissue. First, the individual units obey the direction of the excited normal mode, i.e., they resist attempts at removing their oscillations from that directed by amplitude and phase of the normal mode. Second, in regions in which the electric field created by the normal mode is inhomogeneous, forces are exerted on oscillating individual units. This latter may be considered as a generalization of the case treated in Section IV,B where attraction or repulsion between two equal units was found depending on whether the combined frequency ω_- or ω_+ is excited. In this case a simple result arose when the distance between the two units is larger than their linear dimensions. For then only giant dipole oscillations of single units must be considered. In our present case the situation is much more complex if we are interested in the force between a selected single unit and all the others forming, e.g., a tissue. For the electric field due to the vibrating system of many units may be relatively strong at distances much larger than the size of a single unit even if the system as a whole does not carry out giant dipole oscillations—the latter would be prominent only at distances larger than the linear size of the whole system. Clearly we have to distinguish two types of forces: (1) those exerted by the whole system (whose ω_b mode

is strongly excited) on a single unit outside it at a distance larger than the linear size of a single unit; (2) forces acting within the whole system, which together with standard short-range forces determine the mean distance between units, and possibly also the macroscopic shape. In analogy with the results found in Section IV,B, we expect attraction when $\omega_b < \omega$, i.e., in the case of mode softening. It should be mentioned in this context that since the band of polarization waves ranges from $\omega_1 < \omega$ to $\omega_2 > \omega$, and since its width must increase as the mean distance between units decreases, ω_b is expected to decrease with decreasing mean distance when $\omega_b < \omega$ so that an attraction between units arises on excitation of ω_b . Hence a mean distance is established through conventional short-range interaction. One would also expect that, for similar reasons, individual units of the same kind are attracted to the bulk when they are separated from it. Such attractions may be strongly anisotropic if the excited normal mode possesses a highly anisotropic field. A crude example would be a narrow, long cylinder with a giant dipole oscillation along its excited axis. Relatively high fields, i.e., strong attraction, will then exist near its poles only. Something quite similar holds for a sheet of finite size, where all units oscillate in phase perpendicular to the main surface. If, on the other hand, neighboring units oscillate in opposite direction then their fields cancel to a first approximation. At the same time fields near the edges may be relatively strong.

Clearly, much weaker forces are expected to be exerted on units whose frequency ω' differs sufficiently from ω , corresponding to the case of two units with different frequencies, discussed in Section IV,B.

Considering the units as cells of a tissue it should be remembered that frequencies of greatly differing orders of magnitude can be expected as discussed in Sections III,A and IV,A. Modes with such different frequencies are based on different degrees of freedom and do not interact strongly. As a consequence it is feasible that quite a number of different frequencies are excited simultaneously in a cell.

The second type of consequence from coherent excitation of a certain mode ω_b refers to the enforcement of the amplitude x in a single unit by the excited mode, as follows from Eq. (75). In principle this may have far-reaching influences on processes that are nonlinearly coupled to the x vibration. Consider, for instance, that this vibration is carried by the nucleus of cells and that initiation of certain processes preceding cell division requires reduction of x vibration, e.g., to thermal magnitudes. Then if a particular cell is a part of a tissue in which ω_b is strongly excited, so that x satisfies (70), the required reduction of amplitude must work against its stabilization, which follows from (75). In this model, the stabilization of the amplitude of single cells thus acts against the processes leading to cell

division. A similar argument might hold for mutation. Clearly more explicit considerations can be developed; to be realistic, however, they require more detailed experimental knowledge than is available at present, but it should be mentioned that a very great number of possibilities exist, depending on the spatial structure of the particular normal mode. Thus if we consider a sheet with in-phase oscillations perpendicular to the surface but with a boundary condition of vanishing amplitude at the edges, then the inhibition of cell division would not hold for cells at the edges.

Thus the two consequences of coherent excitation of certain modes in an extensive system such as a tissue may lead to (1) selective attraction or repulsion of individual units, and (2) stabilization of certain oscillations, which can be overcome only by a supply of sufficient energy. It must be stressed, at this point, that the coherent excitation also requires a supply of metabolic energy. A competition between this and the consequences of (2) may thus arise.

To turn now to the cancer (Fröhlich, 1977a, 1978) problem it must be mentioned first of all that in a normal tissue, cells are subject to a control with regard to cell division and to the occurrence of mutations. This control is absent in cancer, and the basic problem thus is to find the origin of the control in normal tissues. Now it has been shown above that a wide range of controls can be exerted by the excitation of coherent vibrations extending over certain regions of a tissue. This may lead to the attraction of cells with equal frequency to certain selected regions. Particular frequencies could thus be characteristic for particular cell differentiation, as first proposed by F. Fröhlich (1973).

The excitation may also lead to the inhibition of cell division, and of the occurrence of certain mutations, if these can occur only when the amplitude of the coherent vibration in a particular cell has been reduced. Since both establishment of the coherent vibration and of cell division with or without mutation require metabolic energy, it is a matter for detailed specification of a model to decide to what extent the control can be effective.

E. Brain Waves

In recent years it has been found that electric fields at very low intensity in the region of 10–20 Hz as well as in the microwave region, appropriately modulated, can severely influence the electroencephalogram (EEG, brain waves) as well as calcium Ca^{2+} efflux and other brain activities (as discussed by Adey and Bawin, 1977). It has also been reported by Elul (1974) that during certain periods large regions oscillate coherently.

It is possible, in principle, to account for this on the basis of our general model of coherent excitations in terms of collective enzyme activities (Fröhlich, 1977c).

This possibility arises within the framework of the "greater membrane" of Schmitt and Samson (1969), which is shown to contain globular proteins that can act as enzymes. As pointed out in Section IV,B, it has been conjectured that the active state of enzymes possesses the high electrical polarization discussed in Section II,B. Thus consecutive excitation and deexcitation of enzymes will lead to electric polarization and depolarization, i.e., to an electric vibration whose frequency will be determined by the rate of the relevant chemical reactions.

Consider now a system of enzymes and their substrates, excited through metabolic processes to our coherent vibrations, which as shown in Section IV,B may lead to selective long-range attraction. Let N be the number of excited, Z the number of nonexcited enzymes, and S the number of substrate molecules in a certain region. Then

$$dN/dt = \alpha NZS - \beta N \quad (76)$$

$$dS/dt = -\alpha NZS + \gamma S \quad (77)$$

Here the α term describes the rate of increase of excited enzymes being proportional to Z , the number of nonexcited ones, and to NS , the rate of reactions. Each such process leads to the destruction of a substrate. The term βN describes the rate of spontaneous recombination of excited enzymes, and the term γS represents the rate of increase in the number of substrates due to our long-range dynamic attraction. Furthermore, we assume that the number Z of unexcited enzymes is always in equilibrium, i.e., time independent; α , β , and γ are constant parameters.

The above equations then form the well-known Lotka-Volterra equations, which have oscillating solutions, i.e., we find the required oscillating chemical reactions. We note the time-independent solutions N_0 , S_0 ,

$$N_0 = \gamma/\alpha Z, \quad S_0 = \beta/\alpha Z \quad (78)$$

so that with

$$N = N_0 + \nu, \quad S = S_0 + \sigma \quad (79)$$

we find

$$\dot{\nu} = \gamma\sigma + \alpha Z\sigma\nu, \quad \dot{\sigma} = -\beta\nu - \alpha Z\sigma\nu \quad (80)$$

Thus if we can neglect the quadratic term $\sigma\nu$ then we find harmonic solutions with frequency $(\gamma\beta)^{1/2}$.

We assume now that the enzymes are arranged spatially in a crudely ordered fashion such that their polarizations, when in the excited state, are nearly parallel. The total polarization P of the region is then proportional to ν if we assume that the contribution of the time-independent part N_0 is screened through mobile ions. Furthermore, the time-dependent polarization P induces an electric current whose resistance yields a negative contribution to $\dot{P}$ proportional to P . Also the interaction between excited enzymes may lead to a tendency toward a ferroelectric state and hence to a positive contribution to $\dot{P}$, proportional to P and to a transition probability. We shall make the ad hoc assumption that this transition probability is governed by a (negative) activation energy proportional to $-(P_{\max}^2 - P^2)$, i.e., by a tendency to "fall" into the ferroelectric state whose polarization is $P_{\max}$. This implies that further terms $\dot{\nu}_1$ have to be added to $\dot{\nu}$, where $\dot{\nu}_1$ has the form

$$\dot{\nu}_1 = (c^2 e^{-\Gamma^2 \nu^2} - d^2) \nu \quad (81)$$

Here c^2 , d^2 , and Γ^2 are positive constants. Thus

$$\dot{\nu} = \gamma \sigma + (c^2 e^{-\Gamma^2 \nu^2} - d^2) \nu + \alpha Z \sigma \nu \quad (82)$$

We note that if we neglect the $\sigma \nu$ terms and develop the exponential up to Γ^2 terms, then using the $\dot{\sigma}$ expression of (80) yields

$$\ddot{\nu} = -\beta \gamma \nu + [(c^2 - d^2) - 3\Gamma^2 c^2 \nu^2] \dot{\nu} \quad (83)$$

i.e., a limit cycle without external field [cf. (70)] provided $c^2 > d^2$. If on the other hand $d^2 > c^2$, then the oscillation is damped out through too large an electrical resistance.

Thus in our model, excitation of coherent vibrations in enzymes and substrates leads to long-range interaction and hence to collective enzymatic processes that oscillate with frequency $(\beta \gamma)^{1/2}$, and, as a consequence, to coherent electric vibrations with the same frequency determined by the rate of spontaneous recombination of an activated enzyme (through β), and the rate of long-range attraction of substrates (through γ). Neither value is known at present, but they might well yield a correct magnitude for brain waves, i.e., an order of 20 Hz. The coherent vibrations leading to the long-range interaction on the other hand should be in one of the regions discussed in Section III,A, i.e., probably between 10^9 and 10^{13} Hz, although lower frequencies are feasible since mode softening might be relevant.

Apart from representing a model for EEG in terms of collective chemical processes, Eq. (82) has other far-reaching consequences, which have been discussed by Kaiser (1977a,b, 1978a,b) but are beyond the scope of this chapter.

It would now be necessary, however, to extend the discussion to the case of more than one type of enzymatic processes. They all will then contribute to the total electric polarization P , which thus must yield an interaction between these processes, and it will have to be investigated whether or not they can coexist. It is feasible then that weak external fields at a frequency $(\beta\gamma)^{1/2}$ can have far reaching consequences on the selection of one or another limit cycle. In the present context it must be pointed out that this again would contribute a trigger effect, the main energy involved being available in the relevant collective chemical processes.

In addition, radiation in a frequency region that influences the coherent electric vibrations resulting in long-range interaction (e.g., microwaves or millimeter waves) will interfere with this interaction and hence with the subsequent processes including the EEG.

V. EXPERIMENTS

A. General

It is one of the most general biological features that energy supplied in terms of food, or of sunlight, is in part used to build up and maintain a very complex organization. A plant might well get warm under the influence of intense sunlight; yet no one would dream of suggesting that in this case the action of the sunlight is entirely thermal (photosynthesis!).

This type of argument is, however, one that has a large following in the discussion of the biological effects of microwaves. It should be obvious, therefore, that it is quite impossible to classify biological effects of microwaves in terms of their intensities alone as has been suggested, e.g., in the "Conclusions" of the International Symposium at Warsaw in 1973; for the absorbed energy depends on the dielectric properties of the particular material, which varies with the frequency of the microwave and depends on the composition of the particular system; moreover as indicated above, nonthermal effects may well be accompanied by general heating. For a decision it would be necessary to heat the system by different means and compare the resultant effects with those found by microwave heating. Such a procedure might, however, give rise to misunderstandings, for not only could one imagine the occurrence of "hot spots" characterized by a subregion with high absorption in conjunction with low thermal conductivity, but also it might occur that certain degrees of freedom are only loosely coupled with the rest of the system. If these interact strongly with each other then they may be endowed with a tem-

perature different from that of the rest of the system. As mentioned before, effects of this kind are well known in physics, e.g., hot electrons. We should agree, therefore, to define an effect as thermal when it can be produced by ordinary heating. We should also agree that it is impossible to establish a general intensity limit below which no biological effects occur. Such limits must depend on frequency and on the particular biological system.

We may also discard here direct effects of the electric field of the microwave on the biological system, as this would require unrealistically high intensities. Any change of a system under the influence of electric fields leads beyond the realm of linear response. For most molecules this is well known to be beyond 10^5 V/cm. Biological systems may, however, be very "soft" in this respect. It must be observed, however, that changes in the constitution of a system by an electric field must arise from a displacement of charges such that their potential energy changes by more than kT . We note that for the electronic charge e this implies a change of electric potential by $V = kT/e = 3 \times 10^{-2}$ V. For a displacement by 10^{-6} cm (thickness of a membrane), the corresponding electric field is $E \approx 3 \times 10^4$ V/cm. Since the intensity of an electromagnetic wave in vacuum is $E^2 c / 4\pi$, the above field is carried by a wave of 10^6 W/cm². The field changes, of course, as the wave enters a dielectric medium with complex structure. We may mention in this context the estimate by Schwan (1974), who finds that in a biological membrane an alternating membrane potential of only 1.5×10^{-7} V is induced by a 3-GHz microwave of intensity 10 mW/cm². To obtain the above-mentioned 3×10^{-2} V thus requires an increase in intensity by (neglecting the dielectric constant) a factor $(2 \times 10^5)^2$ leading close to the above estimate.

From these considerations it seems fairly safe to conclude that microwave effects on biological systems that cannot be explained in terms of heating must arise from trigger effects as treated in Sections IV,C and IV,E. It may be remembered that in the models treated in these sections the principal energy required in the relevant processes is stored by the biological system; the role of the microwave consists in triggering the occurrence of the particular effect. A detailed mathematical treatment would require a time-dependent solution of Eq. (23) when the microwave supply of energy s_m [cf. Eq. (64)] is switched on at a time at which the biological energy supply s_B was below the critical s_0 . It can be seen qualitatively, however, that as s_m is switched on, the system will accumulate more energy (supplied by microwaves) than it loses, until after a period, which might be considerable, Bose condensation is achieved, provided $s_m > s_0 - s_B$. Such systems, clearly, are subject to great fluctuations, and appropriate treatment must therefore incorporate these effects.

According to this theory, as discussed in Section III,C, the microwave may be absorbed in any level of the relevant band as most of the absorbed energy will be channeled into the mode that is to be excited. One may therefore expect the same biological process to be triggered through a whole range of microwave frequencies. It may also be contemplated that the strongly excited mode undergoes a further transition, similar to that discussed in Section III,B.

In the particular model discussed in Section IV,C it was assumed that the biological event observed to be triggered by microwaves consists of a whole chain of steps of which only the first one depends on the coherent excitation of the relevant level. This first step takes a time t_1 , which decreases with increasing excitation, i.e., with increasing rate s_m of supply of microwave energy; all the other steps take a time t_2 independent of s_m . As a consequence, the total time $t_1 + t_2$ tends toward the s_m -independent value t_2 as the microwave intensity increases. In this form this result, however, holds only if the microwave intensity is the same at all relevant systems. If, on the other hand, the relevant systems are distributed such that owing to absorption the microwave intensity differs, then under rather general circumstances a logarithmic increase with s_m precedes the independence of it. Consider, for instance, a homogeneous distribution of systems in a layer of thickness L with the microwaves impinging perpendicularly to the surface. The rate at which the biological process proceeds now depends on x through $s_m(x)$. Thus (63) using (64) must now be replaced by

$$B(x) = \frac{1}{t_2 c + [s_m(x) - \Delta]^n}, \quad \text{if } \Delta > 0, \quad s_m(x) > \Delta \quad (84)$$

$$= 0, \quad \text{if } s_m(x) < \Delta \quad (85)$$

Here $c = a/t_2$ and $\Delta = s_0 - s_B > 0$ are independent of x .

The observed mean rate $\bar{B}$ is then obtained as

$$\bar{B} = L^{-1} \int_0^L B(x) dx \quad (86)$$

and it has been assumed that the biological rate of supply of energy, s_B , is below the critical value s_0 .

We carry out the integration of (86) for the case $n = 1$, assuming

$$s_m(x) = s_m e^{-\alpha x} = z \quad (87)$$

Define a value $x = d$ by

$$s_m(d) = s_m e^{-\alpha d} = \Delta \quad (88)$$

such that $B(x) = 0$ for $x > d$. Then

$$\begin{aligned}\bar{B} &= \frac{1}{\alpha L t_2} \int_{z_0}^{z_1} \frac{z - \Delta}{z - \Delta + c} \frac{dz}{z} \\ &= \frac{1}{\alpha L t_2} \left(\log \frac{z_1}{z_0} - \frac{c}{c - \Delta} \log \frac{z_1 z_0 + c - \Delta}{z_0 z_1 + c - \Delta} \right)\end{aligned}\quad (89)$$

where

$$z_1 = s_m > \Delta, \quad z_0 = \Delta, \quad \text{if } L > d \quad (90)$$

so that

$$\bar{B} = \frac{1}{\alpha L t_2} \left[\log \frac{s_m}{\Delta} + \frac{c}{c - \Delta} \log \frac{\Delta}{c} + \frac{c}{c - \Delta} \log \left(1 + \frac{c - \Delta}{s_m} \right) \right] \quad (91)$$

and

$$\bar{B} = 0, \quad \text{if } s_m \leq \Delta \quad (92)$$

Developing for small $s_m - \Delta$, clearly

$$\bar{B} \approx \frac{1}{\alpha L t_2} \frac{1}{2} \frac{\Delta}{c} \left(\frac{s_m}{\Delta} - 1 \right)^2, \quad \text{if } s_m - \Delta \ll \Delta \quad (93)$$

whereas for large s_m

$$\bar{B} \approx \frac{1}{\alpha L t_2} \left(\frac{c}{c - \Delta} \log \frac{\Delta}{c} + \log \frac{s_m}{\Delta} + \frac{c}{s_m} \right) \quad (94)$$

If on the other hand $L < d$, then z_0 must be replaced by

$$z_0 = s_m e^{-\alpha L} > \Delta, \quad \text{or} \quad s_m > \Delta e^{\alpha L} \quad (95)$$

yielding

$$\bar{B} = \frac{1}{t_2} \left(1 - \frac{1}{\alpha L} \frac{c}{c - \Delta} \log \frac{s_m + (c - \Delta)e^{\alpha L}}{s_m + c - \Delta} \right) \quad (96)$$

which for sufficiently large s_m tends toward $1/t_2$, independent of s_m . Thus ultimately the rate $\bar{B}$ will become independent of s_m even for thick layers.

It must be realized, however, that large s_m may lead to nonnegligible heating. If the biological effect on which the microwave acts depends on temperature then this may mask the original effect. The amount of heating is determined by the overall dielectric properties of the material and in contrast to our specific effects may be expected to vary only slowly with frequency. From the point of view of physics, it is thus imperative to investigate the dependence of a particular biological effect on both frequency and intensity of the microwave.

Furthermore, on the basis of our model, the microwave has supple-

mented the amount s_B of biological pumping (i.e., rate of energy supply) by s_m , and thus has triggered a biological effect that, without the microwave, was due to occur at a later stage. In certain cases it might happen that the opportunity for this action occurs only at certain stages of biological development. Whether this retards or accelerates the overall biological development may be different for different cases. It is a most important experimental task to try to specify the particular biological event that is influenced nonthermally by microwaves in a certain frequency region. It must also be emphasized again that if the biological energy supply s_B exceeds the critical s_0 sufficiently, then no effect of microwave irradiation will occur.

Biological developments usually lead to structural changes of systems (e.g., cell division or formation of proteins) and thus are, from the point of view of physics, highly nonlinear effects. This also holds of the nonthermal action of microwaves in our model. It would be wrong, however, to conclude that investigation of fractions of the system, e.g., particular molecules, would exhibit nonlinear response to microwaves, for only the whole active biological system shows such properties.

It may also be tempting to find the frequency regions that show active biological response by measurement of the absorption of microwaves. While this cannot be excluded in principle, it must be realized that the relevant spatial regions might be very small such that the energy absorbed by them is negligible compared with the overall absorption.

Experimental evidence on the frequency and intensity dependence of biological effects of microwaves is, at present, available in the millimeter wave region only and is discussed in Section V,B. This evidence does support the main principles of our model. Trigger action of microwaves is, of course, only one feature of this model, which predicts strong excitation of polar modes through biological pumping. Evidence for this in a higher-frequency region is now accumulating with the use of Raman effect and will be discussed in Section V,C.

The very great number of investigations of the action of 2450-MHz microwaves on animals frequently are carried out at few intensities only, and, therefore, in the absence of frequency variation do not permit, at present, analysis from our point of view. Certain of them show, however, effects at such low intensities that they do give evidence for nonthermal action. Thus a great number of obviously nonthermal effects are described in the book by Baranski and Czernski (1976). While in their present stage of development, most of these effects are beyond the scope of this chapter, the frequently observed difference between pulsed or modulated waves and continuous waves at equal mean intensity should be discussed briefly.

An obvious reason may arise from the nonlinear reaction to microwaves, as illustrated in Fig. 1, in conjunction with Eqs. (63) and (64). Thus, if the microwave energy supply s_m , together with the biological supply s_B , is less than the critical s_0 , then no effect can be expected. For pulsed waves, however, the maximum s_m will be larger than the mean and together with s_B may exceed the critical s_0 . This possibility may be tested by varying the intensity of the microwaves, or by varying the pulse shape and pulse interval.

Another possibility exists, however, namely, that electric fields at the (low) pulse frequency cause an independent effect that acts jointly with effects of the microwave. Clearly it will be necessary then to investigate the effect of low-frequency electric fields alone.

This, in fact, has been established in the case of the action of 147 and 450 MHz microwaves at intensities of 0.1 and 1.0 mW/cm² on the brain. In a summary article by Bawin *et al.* (1978) the effects of extremely low frequency fields (ELF) at frequencies between 6 and 12 Hz and gradients in air of 0.1–0.5 V/cm on the efflux of calcium from cerebral tissue of chick and cat are described. The authors find decreases in the efflux of 12–15% compared with the unirradiated control. The microwave alone showed no effect. Modulation of the 147-MHz wave with frequencies between 6 and 20 Hz, however, showed an increase of the efflux of up to 15%, and modulation of the 450-MHz wave with 16 Hz also showed an increase. Thus the microwave reverses the effect of the ELF fields although alone it exhibits no such effect. This need not mean that the microwave does not cause other effects. Thus on the basis of the model sketched in Section IV,E one could expect that microwaves in a certain frequency region affect the excitation yielding the long-range interaction between enzymes and substrates. This then led to the periodic chemical reactions presumed, in this model, to be the basis for the EEG. It would then have to be shown that the EEG is influenced by the microwave and that it is directly involved in the binding of calcium ions which thus could be affected by changes in the EEG. For it must be realized, in view of thermal fluctuations, that the binding energy of each single calcium ion must be larger than kT . The low magnitude of the applied field that can modify the efflux thus points to a collective effect in the binding.

Concerning the influence of microwaves on the EEG, it should also be mentioned that investigations by Takashima *et al.* (1979) on the effect of 1–10 MHz irradiation modulated by 15 Hz on mammalian EEGs prove the existence of modifications in the case of chronic irradiation of non-thermal level. Acute irradiations, on the other hand, did not produce changes in the EEG.

From the point of view of our theory, as mentioned above, it seems

likely that the EEG is based on the existence of long-range interactions that can be modified by weak external fields. More experimental material will be required in order to make more specific connections. It can be stated with confidence, however, that collective effects must be prevalent in all phenomena that can be dealt with by our general theory, and that the majority of them will refer to collective vibrations. For this reason it seems desirable to support the existence of such vibrations directly by experimental work.

Experiments in this direction have been devised for the case that involves protons in the vibrational mode, e.g., by Palma (1973), Aiello *et al.* (1973), and Vento *et al.* (1979). These authors emphasize the difference to be expected between independent and collective properties when protons involved in a collective mode are replaced by the twice as heavy deuterons. Their results so far support the existence of collective excitations in a water–agar system. This in turn touches on the possible properties of cell water (cf. Drost-Hansen and Clegg, 1979), assumed to exist in a number of modifications of ordinary water. The task thus exists of finding the relevant frequencies with the possibility of soft modes leading into the microwave regions.

B. Experiments with Millimeter Waves

In the previous section it was noted that direct action of the electric field of microwaves would require unrealistically high intensities. Non-thermal action of microwaves, in particular at 2450 MHz, seems to be established beyond doubt. We must therefore take the attitude outlined in the preceding theoretical sections, although details of particular models may have to be revised in due course. The main general principle, however, should hold, namely, that the microwave acts as a trigger for biological processes. If this is to occur at very low intensities, then the biological system must be able to store the energy supplied by the microwaves. This, in turn, will involve both frequency and time-dependent processes.

The region of millimeter waves is the only microwave region in which at least some of these requirements have been fulfilled and the results, in fact, seem to support the main features of the theory (cf. Fröhlich, 1975b). In order to gain sufficient insight, these requirements must be extended to include biological specification, for quite likely the action of a microwave of given frequency is restricted to certain periods in the development of biological systems. This may, in particular, be so for single cells or for growing tissue, while it need not hold for tissue control as presented in Section IV,D.

1. Influence of Microwaves on *E. coli*

We first discuss the influence of microwaves in the 70–75 GHz region on the rate of growth of *E. coli* bacteria, carried out by Webb and Booth (1969) and by Berteaud *et al.* (1975). The latter use an intensity of 10 mW/mm², probably much larger than the intensity used by Webb and Booth. The irradiation times were three and four hours, respectively. In Fig. 2, N_1 and N_2 are the respective numbers of viable bacteria with and without radiation, counted after a given period of irradiation. Berteaud *et al.* give the accuracy of the cell count as 5%, small compared with the observed effects. Both groups agree that at 70.5 and at 73 GHz the rate of growth is reduced by the irradiation. The difference in magnitude might be connected with the different intensities, for temperature increase in the region of 37°C, used in the experiments, will increase the rate of growth. The magnitude of the reduction of growth may thus have been reduced in the experiment by Berteaud *et al.* through thermal effects.

These authors also find from a number of further experiments that the radiation has no lethal effect on the bacteria and does not influence the

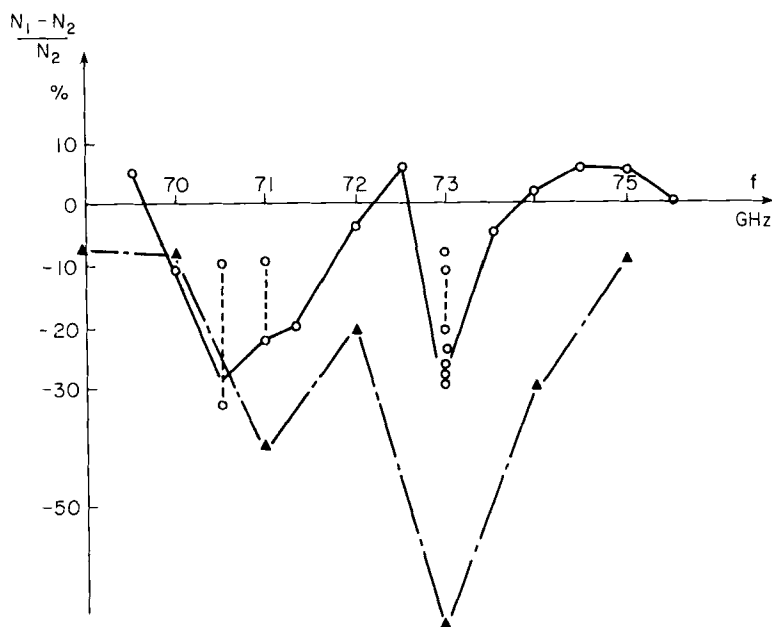


FIG. 2. Growth of *E. coli* bacteria as a function of the frequency of the applied radiation. N_1 is the number of bacteria subjected to radiation, N_2 without radiation (control). O, Results by Berteaud *et al.* (1975); ▲, by Webb and Booth (1969). From Berteaud *et al.* (1975).

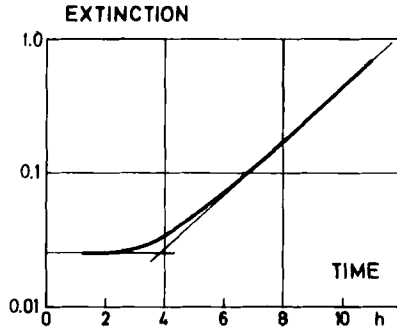


FIG. 3. Optically measured growth of yeast cultures as function of time. The straight line represents exponential growth. Its slope is the rate of growth. From Grundler *et al.* (1977).

rate of mutations. From the latter, in particular, they conclude that the radiation does not act on DNA. This need not imply, however, that the millimeter waves do not influence dynamic processes involving DNA during growth or cell division. In fact, the large effect obtained by Webb and Booth (1969) at very low intensities definitely points toward a trigger effect of the nature described in Section IV,C, although its biological specification would require further investigation.

Such investigation might be of the nature of the experiments carried out by Webb and Dodds (1968) at a different frequency, 136 GHz. The intensity in this case was only $50 \mu\text{W}/\text{cm}^2$. Growing bacteria show a "lag" phase before exponential growth sets in, similar to the case of yeast, illustrated in Fig. 3. The authors find that if the washed cells were immediately exposed for about four hours to microwaves before being incubated in the nutrient, then no cell division took place in a period in which the nonirradiated control multiplied by a factor 6. The radiation was not lethal, however, as the number of viable cells did not decrease. When, on the other hand, the cells were put into a nutrient for 90 min before being exposed to the radiation, then cell growth did take place, although at a very low rate compared with the control.

Clearly it would be most desirable to integrate and extend the two types of experiment, and to measure the dependence of the various effects on intensity.

2. Influence of Millimeter Waves on Yeast Cells

The rate of growth of yeast, like that of bacteria, increases with increasing temperature in the region between 30 and 35°C. The reduction of the rate of growth, which as discussed below arises under the influence

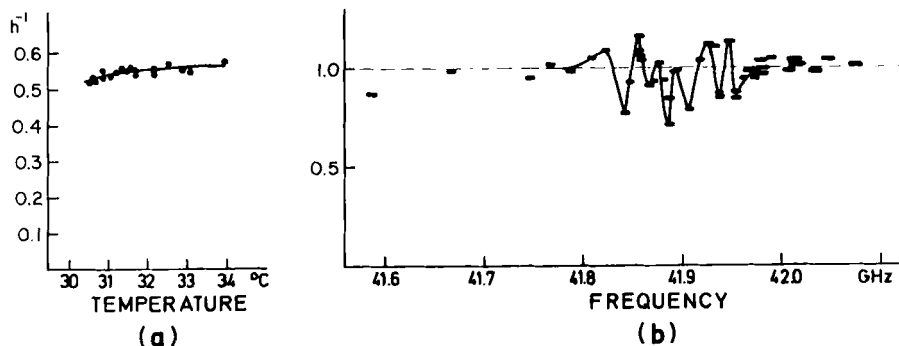


FIG. 4. (a) Temperature dependence of the rate of growth of yeast cultures in the absence of radiation. (b) Rate of growth of yeast cultures with radiation divided by the rate without radiation as a function of frequency.

of millimeter waves at certain frequencies, can therefore not be ascribed to thermal effects.

Devyatkov (1974) finds decreases in total growth of 35, 20, and 40% at 7.16, 7.17, and 7.19 mm, respectively, and an increase of 30% at 7.18 mm. The much more extensive experiments by Grundler *et al.* (1977*; Grundler and Keilmann, 1978) cannot be compared directly with the above, as they use a different genus. The growth of yeast cultures was measured optically as demonstrated in Fig. 3. After a lag period of about three hours the growth becomes strictly exponential as seen in the logarithmic plot. The slope of this section represents the rate of growth, the final result of the measurements. Figure 4a shows the result in the absence of irradiation at various temperatures. The very high degree of reproducibility should be noted. Figure 4b gives the result in the presence of radiation between 41.4 and 41.9 GHz under otherwise identical circumstances. A remarkable set of narrow resonances quite outside the fluctuations seen in Fig. 4a will be observed. The intensity of the radiation was of the order of 2 mW/cm². While this is unlikely to cause significant heating, clearly, as mentioned above, this would increase the rate of growth, whereas decreases have been observed at several frequencies. Dependence on intensity, unfortunately, has not been measured yet, nor have the biological conditions been varied further.

* The experiments on the influence of millimeter waves on the rate of growth of yeast cells (cf. Fig. 4) are at present being repeated by W. Grundler (*J. Collect. Phenom.-Rev.* in print and personal communication) with highly improved accuracy. So far two resonances have been obtained containing ten about equally spaced experimental points within a frequency range of 10 MHz, thus giving very strong support to the previous results. No dependence on the intensity was found as required by the theory, and demonstrated in Fig. 1.

From the point of view of our theory, the resonance frequencies that show a decrease in the rate of growth must be attributed to a band of vibrations, those that show an increase to another overlapping band. The spatial regions to which these two bands belong may, of course, be well separated. The sharpness of the resonances points toward regions sufficiently separated from cell water, because contact with it would yield considerable friction, i.e., line broadening, although this could be overcome by certain nonlinear excitation. According to our theory, the microwave energy supplied to each mode is channeled, through χ processes, to the mode that is strongly excited. Also once this mode is strongly excited, it might undergo further transitions and then be stabilized in an "activated" state. The present investigations will have to be extended considerably in order to be able to investigate these possibilities experimentally.

3. Induction of Colicins

Colicins are a class of proteins manufactured on certain occasions by many bacteria of the *E. coli* group (cf. Luria, 1975). They act as antibiotics that kill cells of bacterial strains related to the strains that makes them. There exist very many kinds of colicins; each is the product of an extrachromosomal gene. Smolyanskaya *et al.* (1974) have shown that the induction of colicin can be significantly influenced by millimeter wave irradiation. This is illustrated in Fig. 5, where the appropriately defined ratio of

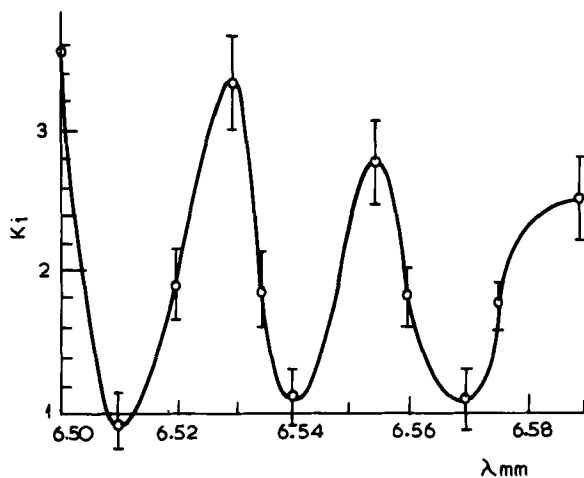


FIG. 5. Induction coefficient K of colicin under the influence of radiation divided by the same without radiation as a function of wavelength. From Smolyanskaya and Vilenskaya (1974).

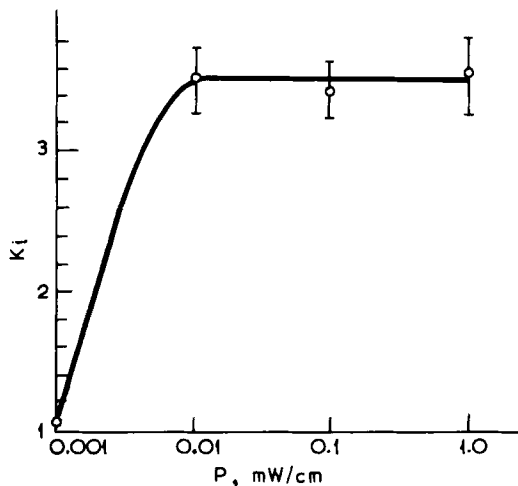


FIG. 6. Dependence of K on the intensity of the radiation. From Smolyanskaya and Vilenetskaya (1974).

colicin induction with and without radiation is presented as a function of wave length. Figure 6 shows the dependence on intensity (probably at 6.5 mm).

It will be noted that no effect is observed at an intensity of $1 \mu\text{W}/\text{cm}^2$, whereas the effect is independent of intensity between $10 \mu\text{W}/\text{cm}^2$ and $1 \text{ mW}/\text{cm}^2$, i.e., over a range of a factor 100. This is exactly as required by theory (cf. Fig. 1). The logarithmic increase [cf. Eq. (93)] would have to be expected to take place between 1 and $10 \mu\text{W}/\text{cm}^2$ within which region no measurements have been performed. Should experiment provide, however, a very sharp transition within this region, then Eq. (63) would apply and it would have to be concluded that only a relatively thin layer, within which the microwave intensity does not change much, is responsive.

Figure 5 again indicates resonance behavior although the experimental points are not sufficiently close to permit comparison of the width of the resonances with that observed in yeast (Fig. 4b). Again we may assume that the microwave energy, absorbed in the resonance region is channeled, through χ processes, to the mode that is strongly excited. Its frequency is not necessarily among those used in the experiments. Again the possibility of a further transition of the excited mode arises. Such processes may, in principle, require considerable time, and the experiments on the dependence of the effect on time, presented in Fig. 7, are of particular interest in this connection. They do, in fact, exhibit the ability of the

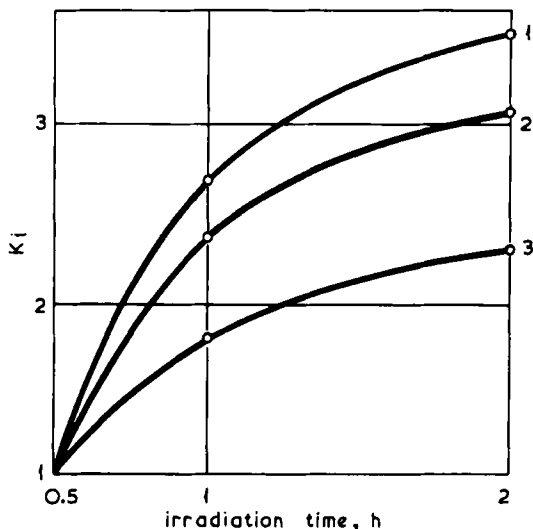


FIG. 7. Dependence of K on the irradiation time, for 6.5 (1), 5.8 (2), and 7.1 (3) mm. From Smolyanskaya and Vilenskaya (1974).

system to store the supplied energy over considerable periods. Considering the low intensity required for the effect, the system must possess such storing capacity to overcome the appreciable noise. The required time should decrease with increasing intensity; experiments in this direction would be desirable.

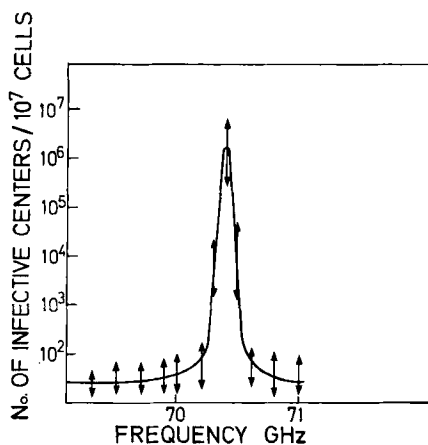


FIG. 8. Frequency dependence of the induction of lambda prophages. From Webb (1979).

Motzkin *et al.* (1979) have also been able to demonstrate an enhancement of colicin induction under irradiation at 5.8 mm.

Related in certain ways to the induction of colicin is the induction of lambda prophages in *E. coli* bacteria. These bacteria frequently contain the DNA of a certain virus. Certain circumstances such as irradiation by X ray or ultraviolet light induce this prophage so that the virus is produced and finally destroys the cell. Webb (1979) has shown that the growth can also be induced by millimeter waves in a narrow frequency region near 70.5 GHz, but not outside it, as shown in Fig. 8. It will be seen that 10^6 out of 10^7 cells are induced whereas only 10 out of 10^7 cells are induced without irradiation, or with irradiation but outside the active frequency region. The effect also depends on intensity as required by our theory as shown in Fig. 9, although the decrease at higher intensities has not yet been explained. It should be mentioned, in this context that the irradiated bacteria are used in a monocellular layer so that Eq. (82) can be applied. By investigation of synchronized cells it has been found that the microwave irradiation is effective only during a narrow period of cell development (cf. Fig. 10).

These results clearly eliminate thermal induction, which would not be restricted to a narrow frequency region. It is further shown that the effect occurs only when certain nutrients are used; it might be, of course, that other nutrients show a similar effect outside the frequency range investigated so far.

It should be pointed out at this stage that effects of the type described above depend on many parameters, some of which, in particular with

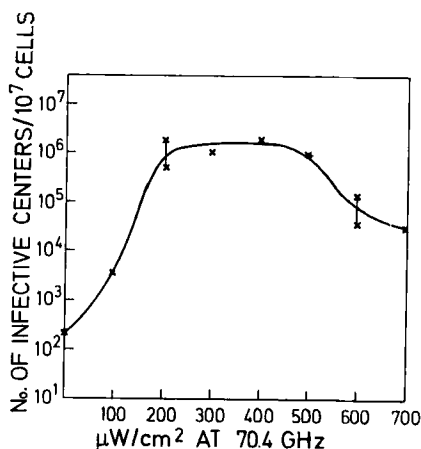


FIG. 9. Dependence of the inductions of lambda prophages on the intensity of radiation. From Webb (1979).

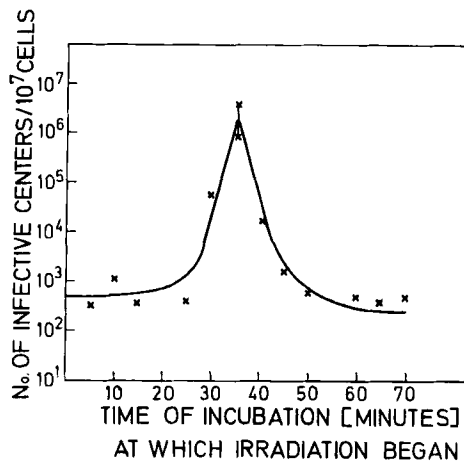


FIG. 10. Dependence of the induction of lambda prophages on the time of incubation of a cell. From Webb (1979).

regard to the biological properties, are not known at present. Thus it has been mentioned to the author at various instances that large effects of the type described above have been confirmed, but have at later stage disappeared. This might well arise from a change of biological specifications that so far have not been brought under control.

4. Influence of Millimeter Waves on Protein Metabolism

An investigation on the influence of 7.2 and 7.6 mm waves on certain aspects of protein metabolism by Manoilov *et al.* (1974) might help toward specification of the action of millimeter waves. These authors have introduced certain fungi and bacteria into nutrient solution and after a certain period have investigated the free amino acid content for the unirradiated control, and the 7.2 and 7.7 mm irradiated case. The intensity was 4–5 mW/cm², the irradiation time 3 hours.

Figure 11 shows for the case of *Staphylococcus* the percentage change of the irradiated samples over or below the control. Clearly several amino acids have been strongly influenced by one of the two frequencies but not by the other. Thus, e.g., in aspartic acid, while the 7.6 mm wave has no influence, the 7.2 mm wave has increased the content by 72%. From the point of view of our theory we should conclude that the 7.6 mm but not the 7.2 mm frequency belong to a band containing a frequency that when excited is involved in the appropriate chemical processes. This might be

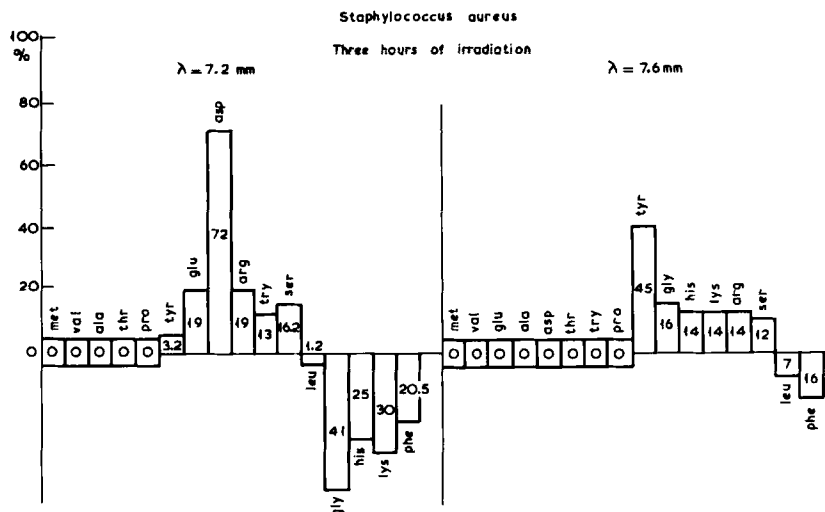


FIG. 11. Influence of radiation on the content of various amino acids in *Staphylococcus*. The percentage change over nonirradiated cases, for two wavelengths, is presented. From Manoilov *et al.* (1974).

connected directly with the action of an enzyme, as the authors suggest. It might, alternatively, be connected with membranes, or other cell constituents, involved in the process. From our discussion of frequencies in Section IV,A, we would be inclined to favor the latter, as enzymes are likely to possess higher frequencies. It must be realized, however, that this statement might have to be modified through the possibility of mode softening.

It should also be mentioned in this context that Tuengler *et al.* (1979) have been unable to observe an influence of microwaves in the frequency region between 40 and 115 GHz (7.5–2.6 mm) on certain other enzymatic function *in vitro*.

Another experiment concerning the influence of microwave irradiation on biosynthesis was carried out by Webb (1975). He investigated the synthesis of DNA and RNA in *E. coli* bacteria using a radioactive tracer method. In no case was the rate of synthesis increased by microwaves. Certain frequencies reduced the rate, however, to about 1/10 of the unirradiated one for DNA, notably 59, 66, 73, 129, 136, and 143 GHz if the cells were grown in a glucose–amino acid nutrient. None of these frequencies influenced RNA synthesis. In a glucose– NH_4^+ medium, however, the relevant frequencies are 61, 66, 71, 76, 126, 136, and 141 GHz.

In terms of our theory, we should conclude that the synthesis is con-

nected with certain vibrations whose frequencies depend on the nutrient. We would suggest that they are based to certain regions in or near membranes in conjunction with nutrient molecules in or near these regions.

5. Influence of Microwaves on Bone Marrow Cells of Mice

Finally in this section the experiments by Sevastyanova and Vilenskaya (1974) must be discussed as investigations on frequency, intensity, and time dependence have been reported. The investigations deal with the damage to the bone marrow of mice arising from exposure to X rays or from the use of certain chemicals. When prior irradiation with microwaves has taken place, then at certain frequencies and intensities the damage is vastly reduced. In Fig. 12 it is shown that while the number of nonirradiated bone marrow cells is reduced to about one-half by exposure to X rays, prior irradiation by microwaves of certain selected frequencies reduces the damage such that about 80% survive. No effect exists, however, at other frequencies.

Figure 13 demonstrates the dependence of this effect on intensity. The effect is absent below 9.9 mW/cm², then steplike reaches its maximum at

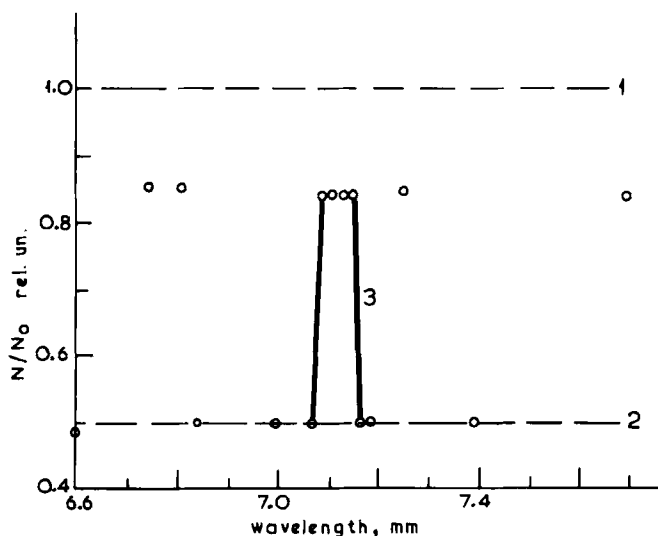


FIG. 12. Dependence of the relative number of bone marrow cells N/N_0 on the wavelength of the radiation. N , number of cells; N_0 , number of cells without radiation. 1, control; 2, X-ray irradiated; 3, X-ray and millimeter-wave irradiated. From Sevastyanova and Vilenskaya (1974).

10 mW/cm², and remains unchanged up to 80 mW/cm². Figure 14 shows the time dependence—it takes 75 min of irradiation to reach the full response.

The authors report that about 70% of the microwave radiation is absorbed in a surface layer of the skin of about 3×10^{-2} cm thickness. Thus for direct action on a deeper region only a very small fraction of the 10 mW/cm² will penetrate. If we take the case of colicin as a guide, then the threshold for action might occur between 10^{-3} and 10^{-2} mW/cm² (cf. Fig. 6) and a reasonable thickness for penetration could arise, especially if we admit the possibility of even lower thresholds. On the basis of our theory the sharp step in Fig. 13 then indicates that a very thin layer only, within which the microwave intensity does not change appreciably, is sensitive to microwaves. For then the local formula (63) can be used without the integration over space carried through in connection with Eqs. (93) and (96). The experimental result then also implies that the first step, which involves the influence of the microwave irradiation, proceeds fast compared with the following steps, i.e., in terms of Section IV,C, $t_1 \ll t_2$.

The frequency dependence presented in Fig. 12, according to our theory, then defines some of the frequencies of the band containing the mode, which will be strongly excited. χ Processes then are responsible for

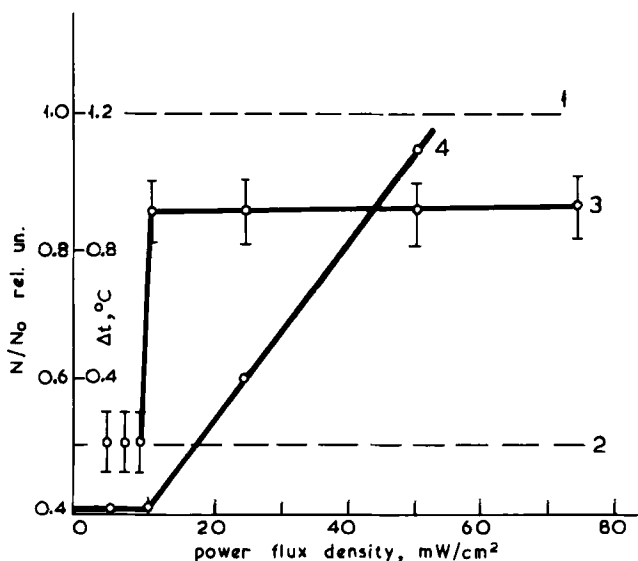


FIG. 13. 1, Control; 2, relative number of X-ray-irradiated bone marrow cells; 3, dependence of X-ray- and millimeter-wave-irradiated cases on the intensity of the millimeter waves; 4, change Δt of skin temperature as function of millimeter wave intensity. From Sevastyanova and Vilenskaya (1974).

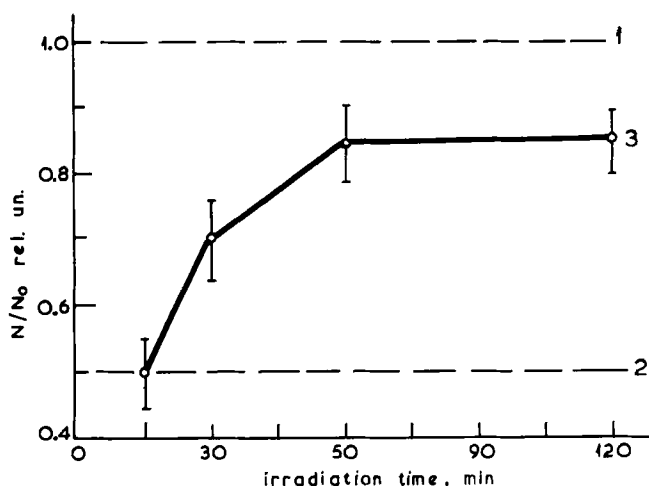


FIG. 14. Dependence of the number of bone marrow cells on the duration of millimeter wave irradiation. 1, Control; 2, X-ray irradiated; 3, X-ray and millimeter-wave irradiated. From Sevastyanova and Vilenskaya (1974).

the channeling of the absorbed energy with this mode. The time required for this follows from Fig. 13. We should expect that this time decreases with increasing intensity and experiments in this direction would shed further light on the above interpretation.

6. Summary

The examples on the action of millimeter waves on biological systems presented in the present section leave no doubt that nonthermal action exists at very low intensities. They give support to the theory presented in previous sections, according to which strongly excited polar vibrational modes play a basic role in the dynamics of biological processes, and that such excitation can be triggered by microwaves.

Measurements of the dependence of various effects on frequency, intensity, and duration of irradiation over an appropriate range are required and when carried out in conjunction with biological specification may lead to new insight. Thus measurement of the dependence on intensity alone, in the above case, led to the conclusion of the existence of a thin layer in which the relevant oscillations should take place.

From the frequency dependence of the rate of growth of yeast we were able to conclude that two different processes must be involved. Measurements on synchronized cells might then help to further specify the relevant biological processes.

Clearly considerable experimental difficulties must be overcome to be able to present a complete picture of the involvement of vibrations in biological processes. Success would, however, have far-reaching repercussions on the understanding of dynamic processes in biology.

C. Raman Effect

According to one of our concepts (Section III,C) metabolic energy excites specific coherent electric vibrations in biological systems at certain stages of its development. The discussions of Section V,B give support to this proposal for frequencies in the millimeter wave region, i.e., near 10^{11} Hz, for only thus is it possible to understand the relatively large biological effects initiated by electromagnetic irradiation in the millimeter region at relatively low intensities. To demonstrate the generality of our concept it is important to give evidence for the existence of relevant excitation in neighboring frequency regions as well. Higher-frequency regions can be reached by Raman effect. It must then be shown that the system possesses proper frequencies in this region, and that these are excited, beyond thermal level, at certain periods of development. For the latter purpose measurement of the intensity ratio of anti-Stokes to Stokes lines can be successful provided the Raman frequency ν is so high that the intensity ratio in thermal equilibrium, $\exp(-h\nu/kT)$, is appreciably smaller than unity. This requires Raman frequencies of the order of 100 cm^{-1} (3×10^{12} Hz) or higher.

At lower frequencies, the ratio of anti-Stokes to Stokes intensities approaches unity already in thermal equilibrium and can thus not be used to demonstrate nonthermal excitation. Should the nonthermal excitation be very strong, however, then measurement of the order of magnitude of the Raman intensity may suffice to establish it.

Systems endowed with proper frequencies in the relevant regions are likely to be based on large molecules, principally proteins, placed inside or outside membranes. In fact, Brown *et al.* (1972) have shown that α -chymotrypsin, an enzyme, possesses Raman frequencies down to 25 cm^{-1} . Genzel *et al.* (1976) also observed at 25 cm^{-1} a Raman frequency of lysozyme when it is crystallized. In solution this line disappeared, probably due to friction in the solution. It may be expected then that the appearance of frequencies of this order in biological systems will depend on the manner in which such molecules are contained in cells. Thus low friction may be expected when they are dissolved in membranes, high friction when floating in cell water. It is important, therefore, to note that Drissler (1980), has shown by Raman effect on algae that certain chlorophyll lines, in the region between 100 and 3000 cm^{-1} are present at low

temperature but disappear at room temperature. Drissler *et al.* (1978), on the other hand, show that lines in the region between 800 and 1500 cm^{-1} , assigned to vibrations of carotenoid molecules (possibly resonating with some chlorophyll lines) are present in life algae. These molecules thus may be built into the cell in a manner characteristically different from that of chlorophyll molecules.

Certain modes in higher-frequency regions have already been discussed in Section III. Apart from this it should be emphasized that biomolecules possess a great number of oscillatory modes in regions above 1000 cm^{-1} , and a great literature exists on this subject. These investigations are of importance for the investigation of structure but have so far not been used to investigate the possibility of collective excitations.

Candidates for collective excitations have been discussed in Section III, and we consider in further detail the results obtained by Biscar and Kollias (1973a,b). These authors show that after careful preparation of poly-L-glutamic acid, Stokes Raman shifts are observed that are an integer multiple of a basic frequency, namely, 950 cm^{-1} when the molecular weight is 93,000 and 860 cm^{-1} when it is 103,000. Shifts up to the sixth harmonic are observed. Since the length of such a molecule is proportional to its molecular weight, the basic frequency is inversely proportional to the length L for small changes of L . This behavior would hold for acoustic modes, but then the frequencies would have to be several orders of magnitudes lower. Electronic acoustic modes, which the authors suggest, i.e., acoustic plasmons, require at least two partly filled energy bands for electrons, a requirement probably not fulfilled in this material. Quasi-free

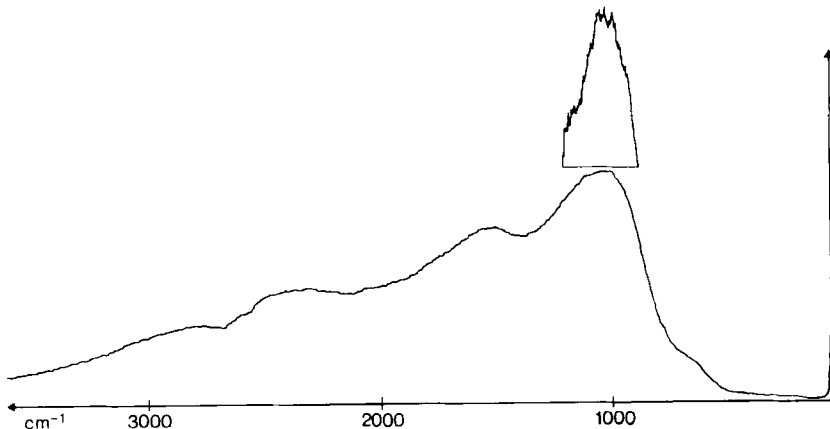


FIG. 15. Stokes Raman spectrum of a paste of *E. coli* bacteria at 7 K. The inset is obtained at higher sensitivity. From Drissler and Webb (1980).

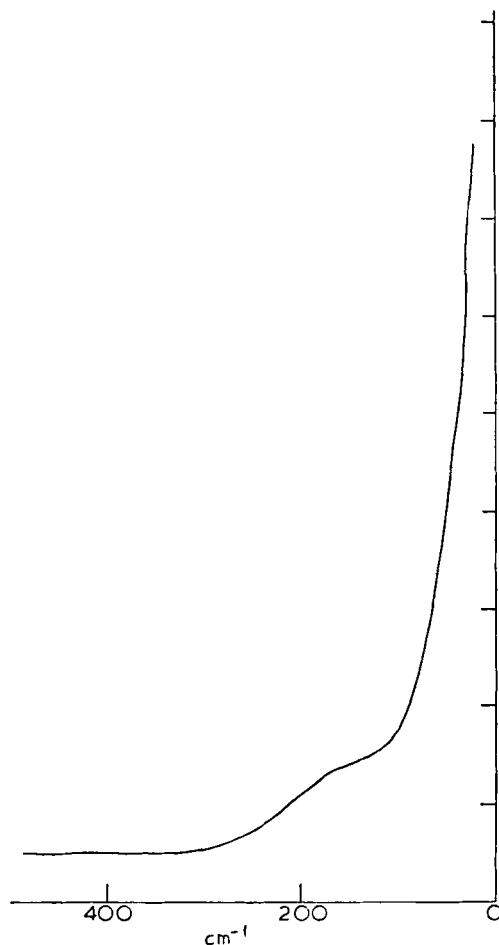


FIG. 16. Stokes Raman spectrum of resting cells of *B megaterium* bacteria. From Webb and Stoneham (1977).

electrons, which according to Kuhn (1949) exist in certain long-chain molecules, might explain the experiments. In this case the electronic energy levels are

$$E = (\hbar^2 2m^*)n^2\pi^2/L^2, \quad n = 1, 2, 3, \dots \quad (97)$$

The basic frequency would then arise from the transition from $n = 1$ to $n = 3$, as the transition $n = 1$ to $n = 2$ is probably prohibited in Raman effect. The harmonics would then correspond to simultaneous excitation of electrons in different molecules. On this assumption the empirical

value of about 10^3 cm^{-1} for the transition yields a length of $L \approx (m/m^*)^2 \times 5 \times 10^{-7} \text{ cm}$, which is not unreasonable (m^* is the effective electronic mass, which would have to be smaller than the electronic mass m).

Turning now to experiments on bacterial cells, Drissler and Webb (1980) find in a densely packed paste of *E. coli* bacteria (about $10^{11} \text{ cells/cm}^3$) at 7 K three broad bands in the regions 900, 1400, and 2000 cm^{-1} . Each band has a width of about 400 cm^{-1} . All three bands possess fine structures of very many lines of width of about 10 cm^{-1} as shown in Fig. 15. This fine structure is well reproducible, and it can be noticed that thermal noise is absent at this temperature.

According to our theory we would identify the broad bands with the bands of modes introduced in Section III. Metabolic activity would then be expected to excite, in living cells, a single line at certain periods of the development of the cells. Each of the broad bands might represent an overlap of several bands of modes whose originating systems (proteins in or outside membranes) might be separated spatially. In living cells we

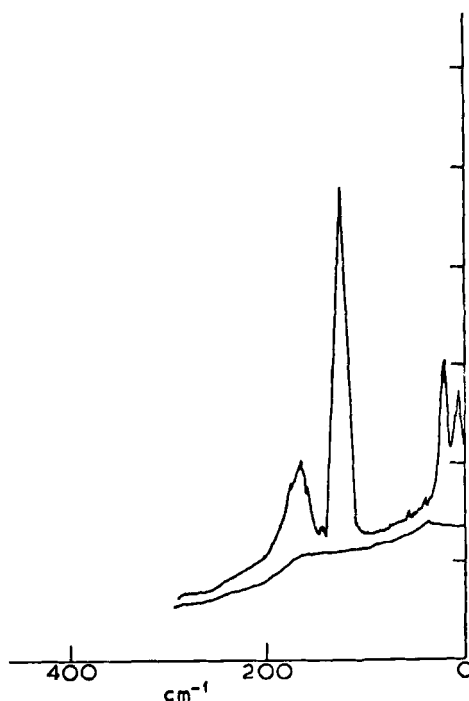


FIG. 17. Same as Fig. 16 with added nutrient. From Webb and Stoneham (1977).

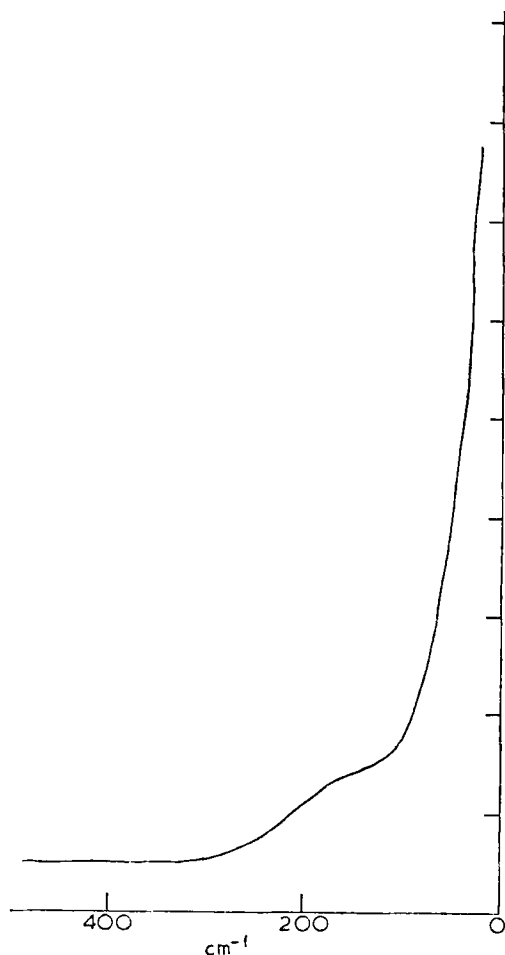


FIG. 18. Raman spectrum of the nutrient. From Webb and Stoneham (1977).

should thus expect the appearance of single Raman lines, both Stokes and anti-Stokes, in the frequency regions of the bands observed at 7 K. In synchronized cells, the actual positions should change with the age of the cells, as does the cell activity.

In fact, earlier measurements as reviewed by Webb (1980) support this picture. In these measurements, unfortunately, Stokes lines only were investigated. It should be observed, however, that the density of cells for *in vivo* measurement is 10^7 cells/cm³ only, i.e., a factor 10^4 below that used in the measurements at 7 K. It seems possible that in thermal equilibrium, the Raman intensity would be too small to be measureable (excluding

resonance Raman effect), and hence that a case for nonthermal excitation can be made although this would have to be supported by more quantitative arguments. It should be mentioned in this context that the intensities of Stokes and anti-Stokes lines are proportional to $(1 + n)$ and n , respectively, where n represents the number of quanta by which a particular vibration is excited. In the frequency region of the bands, the thermal excitation $n_T \ll 1$. It follows that a nonthermal enhancement of the excitation to n quanta requires $n \gg 1$ in order to yield a significant increase of intensity for Stokes lines. For anti-Stokes lines it requires only $n \gg n_T$.

A much simpler picture arises in the case of life algae that were investigated by Drissler and MacFarlane (1978) in a flow system in which an individual cell spends a negligible time in the laser beam. This time is too short to set the photosynthetic process into action so that the results represent the spectral properties (already referred to above) of an *in vivo* cell in the absence of metabolism.

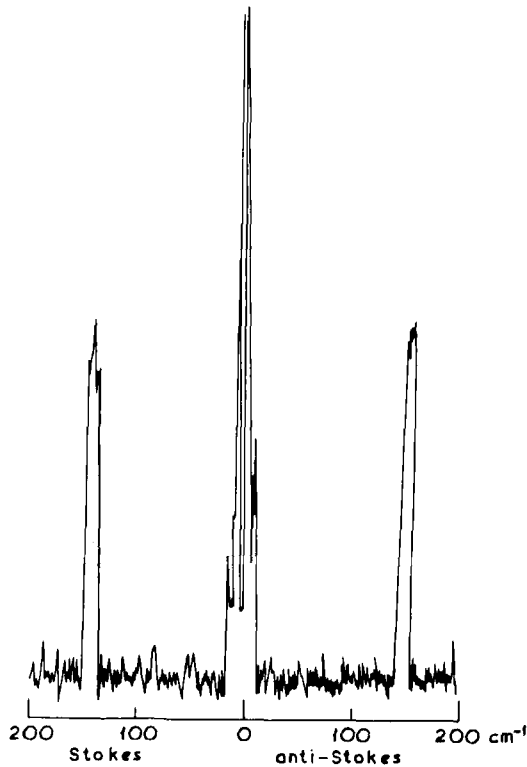


FIG. 19. Stokes and anti-Stokes Raman spectra of synchronized active cells of *E. coli* B bacteria, taken at 40 min after incubation. From Webb *et al.* (1977a).

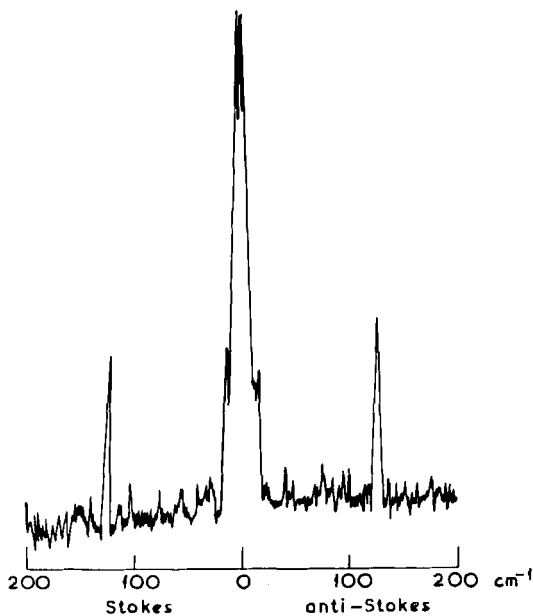


FIG. 20. Stokes and anti-Stokes Raman spectra of synchronized active cells of *E. coli* B bacteria, taken at 50 min after incubation. From Webb *et al.* (1977a).

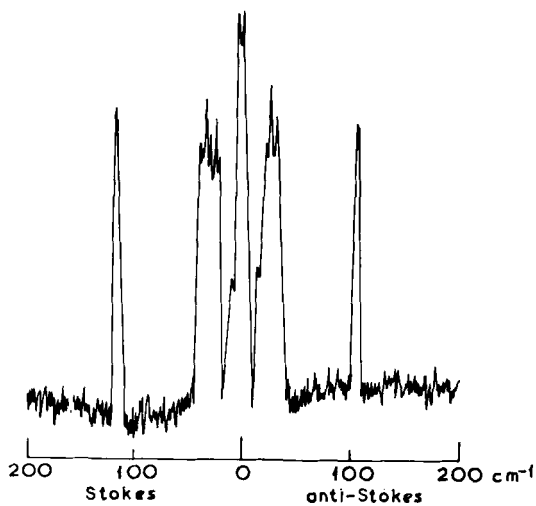


FIG. 21. Stokes and anti-Stokes Raman spectra of synchronized active cells of *E. coli* B bacteria, taken at 60 min after incubation. From Webb *et al.* (1977a).

Addition of white light must then yield the changes arising from the metabolism.

In the region below 200 cm^{-1} measurement by Webb and Stoneham (1977) of Stokes lines on cell of *B megaterium* bacteria at a density of 5×10^7 cells/ml show a broad band only when the cells are resting, but at least two lines when the cells are active and synchronized, as shown in Figs. 16 and 17. The spectrum of the nutrient is shown in Fig. 18. Again a quantitative investigation of the intensities might show that activation beyond thermal excitation is required. The crucial experiment for anti-Stokes lines has been carried out more recently by Webb *et al.* (1977a) on synchronized active cells of *E. coli* B bacteria. Figures 19–21 show the spectra taken 40, 50, and 60 min, respectively, after incubation. We note that the line near 120 cm^{-1} ranges between 118 and 125 cm^{-1} . At these frequencies the ratio of the intensities anti-Stokes : Stokes = $\exp(-h\nu/kT)$ in thermal equilibrium lies between 0.55 and 0.57. Experimentally after

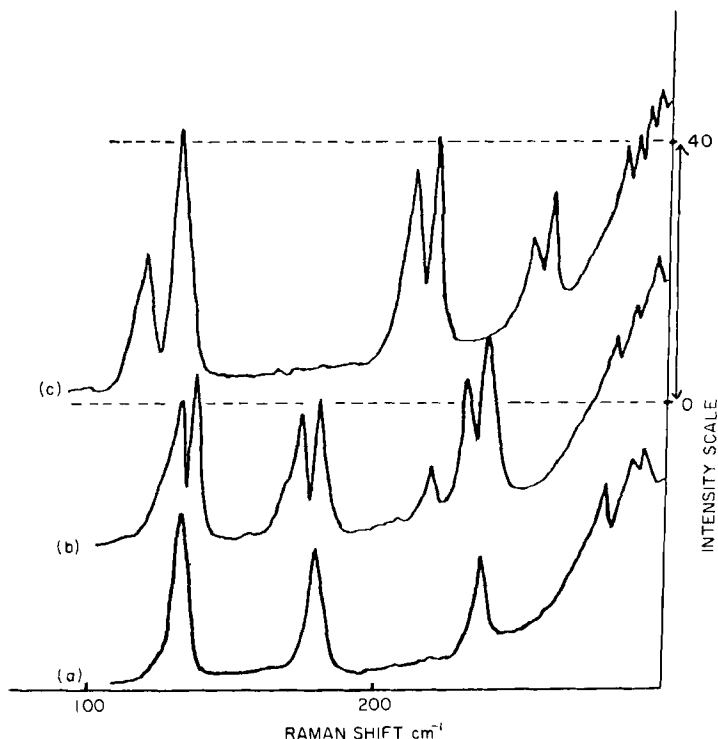


FIG. 22. Stokes Raman spectra of mammary tissue. (a) Normal; (b) "normal" right-side tissue; (c) left-side carcinoma. From Webb *et al.* (1977b). (Reprinted by permission of John Wiley & Sons, Inc.)

averaging over eight spectra one finds 1.01 ± 0.10 , and 0.93 ± 0.13 , respectively. It can be noted that in thermal equilibrium $n_T \approx 1$, whereas at the lower limit 0.8 of this ratio,

$$n/(1+n) = 0.8, \quad \text{or} \quad n = 4 \quad (98)$$

requires a considerable enhancement; using 0.9, this would lead to $n = 9$.

A further set of interesting Stokes lines has been obtained by Webb *et al.* (1977b) from normal mammary tissue, and from tissue of mammary carcinoma, as shown in Fig. 22. Anti-Stokes lines unfortunately are not yet available. The splitting and broadening of the lines of the carcinoma might give support to the theoretical suggestions of Section IV,B, although much more experimental evidence than available now will have to be provided.

Finally it should be emphasized again that certain Raman lines occur at particular stages of development of cells only as shown in the review article by Webb (1980).

VI. CONCLUSIONS

From the experiments on millimeter waves discussed in Section V,B, it can be concluded with reasonable confidence that these waves cause biological effects that can be understood neither in terms of heating nor through direct action of the electric field of the waves. It follows that the electromagnetic wave acts as a trigger to events for which the biological system is already prepared. This implies that the biological system has the possibility of storing the energy of the microwaves until the trigger effect can occur, and in fact we have shown that considerable irradiation times may be required until the effect reaches its full strength.

Such action clearly is of a highly nonlinear nature and therefore cannot be understood in terms of linear response of the system to the impending microwave, using properties of the dielectric function. This latter type of argument has frequently been presented (as, e.g., Stuchly (1979) where, as also seems to be the habit, properties of a dilute system of dipolar molecules only are considered), and it should be added that the frequently propounded idea that the microwave "rotates" dipolar groups is, mildly speaking, misleading. This would require the expression for the dielectric constant to depend on the moment of inertia, which it does not (e.g., Fröhlich, 1958). In addition, it would require quite unrealistically high intensities.

The strongest evidence for the existence of such nonthermal, nonlinear effects arise from their dependence on frequency, intensity, and

time of irradiation. To this we may add the occurrence of large biological effects, in other frequency regions, at very low intensities, although without measurement of the dependence of such effects on frequency the nonthermal nature is not convincing. Thus "microwave hearing" has been found to be a very sensitive temperature effect (e.g., Lin, 1977). One can predict, therefore, that this effect is very insensitive to a variation of the frequency, contrary to the effects discussed in Section V,B.

It follows that an understanding of the nonthermal effects must be based on a general theory of the importance of electric vibrations in biological systems. This theory has been presented in Section III, and its consequences discussed in Section IV. The presentation has been given as general as feasible, consisting largely in the introduction of the concept of collective coherent electric vibrations, and of highly polar metastable states. Both can be excited through metabolic energy (biological pumping). Both can have far-reaching consequences on the dynamic properties of biological systems. It will be required by collaboration between theory and experiment to find the details of these excitations. Clearly microwaves are only one experimental tool in this respect. Others include Raman effects, as discussed in Section V,C.

A new tool, microdielectrophoresis, has recently, in the hands of Pohl (1979), provided new evidence for the existence of electric vibrations of certain rapidly dividing cells. These vibrations occur at relatively low frequencies, and it is of considerable importance to specify the regions that carry out these vibrations.

Solution of this problem, of course, is most essential in all frequency regions that have been discussed here. To this must be added the specification required to find a particular effect. Thus in the experiments by Webb (1979) on the induction of prophages discussed in Section V,B it has been shown that in the investigated frequency region, one particular nutrient only yields the effect, and this particularly strongly in the presence of free oxygen.

Quite generally, insufficient knowledge of specification is likely to be the reason for the difficulty that is sometimes found in reproducing experiments. In this respect, the influence of relatively weak magnetic fields must be mentioned, which, although not understood at present, might have considerable influence on the performance of a number of processes. Thus Shaya and Smith (1977) warn against the use of magnetic stirrers, as treatments by magnetic fields were found to influence the activity of the enzyme lysozyme (*in vitro*). Ahmed *et al.* (1980) also find unexpected effects of magnetic fields on the deposit of lysozyme on glass surfaces, in the presence of oxygen.

A most important requirement arising from the theory is the need for

experiments *in vivo*, for both the basic concepts are connected with excitations through metabolic energy. This does not mean that investigations of metabolically nonactive systems are without value. Thus Raman effects on bacteria at 7 K (cf. Fig. 15) have exhibited the existence of bands of modes that form the structural background of the theory. Metabolic energy will then, at certain stages of development, lead to the coherent excitation of single levels, and evidence for this is also available.

A most important question, which in no case has been answered, refers to the molecular specification and placement to which various excitations are due. In previous sections we have tended to attribute excitations in the 1000 cm^{-1} region to enzymes, the millimeter wave region to sections in membranes, modified by dissolved proteins, and possibly to DNA. The possibility exists, however, that millimeter waves cause transitions between the levels of the band exhibited in Fig. 15. This would involve nonlinear effects and also a slight modification of the theory as such effects are not possible without prior excitation of these levels.

Thus in conclusion it should be emphasized again that microwave experiments have indicated the existence of coherent electric vibrations in biological systems. To elucidate their role it is imperative to investigate the same systems with other techniques such as the Raman effect.

REFERENCES

- Adey, W. R., and Bawin, M. (1977). *Neuroscience Res. Program. Bull.* **15**.
 Ahmed, N. A. G., Norman, M., and Smith, C. W. (1980). *J. Collect. Phenom.* (in press).
 Aiello, G., Micciancio-Giammarinaro, M. S., Palma-Vittorelli, M. B., and Palma, M. U. (1973). In "Cooperative Phenomena" (H. Haken and M. Wagner, eds.), p. 395. Springer-Verlag, Berlin and New York.
 Baranski, S., and Czerski, P. (1976). "Biological Effects of Microwaves." Dowden, Hutchinson & Ross, Inc., Stroudsburg, Pennsylvania.
 Bawin, S. M., Sheppard, A., and Adey, W. R. (1978). *Biochem. Bioenerg.* **5**, 67.
 Berteaud, A. J., Dardalhon, M., Rebeyrotte, N., and Averbeck, M. D. (1975). *C. R. Hebd. Seances Acad. Sci., Ser. D* **281**, 843.
 Bhaumik, D., Bhaumik, K., and Dutta-Roy, B. (1976). *Phys. Lett. A* **59**, 77.
 Bhaumik, D., Dutta-Roy, B., and Lahiri, A. (1978). *Phys. Lett. A* **68**, 131.
 Biscar, J. P., and Kollias, N. (1973a). *Phys. Lett. A* **45**, 189.
 Biscar, J. P., and Kollias, N. (1973b). *Phys. Lett.* **45A**, 191.
 Bilz, H., Büttner, H., and Fröhlich, H. (1980). *Z. Naturforsch. Teil C* (in press).
 Brill, A. S. (1978). *Biophys. J.* **4**, 139.
 Brown, K. G., Erfurth, S. C., Small, E. W., and Peticolas, W. L. (1972). *Proc. Natl. Acad. Sci. U.S.A.* **69**, 1467.
 Bullok, T. H. (1977). *Neurosci. Res. Program, Bull.* **15**, 17.
 Careri, G. (1969). In "Theoretical Physics and Biology" (M. Marois, ed.), p. 55. North-Holland Publ., Amsterdam.

- Careri, G. (1973). In "Cooperative Phenomena" (H. Haken and M. Wagner, eds.), p. 391. Springer-Verlag, Berlin and New York.
- Devyatkov, N. D. (1974). *Sov. Phys. — Usp. (Engl. Transl.)* **16**, 568.
- Drissler, F. (1980). *Phys. Lett. A* (in press).
- Drissler, F., and MacFarlane, R. M. (1978). *Phys. Lett. A* **69**, 65.
- Drissler, F., and Webb, S. J. (1980). *Z. Naturforsch. Teil C* (in press).
- Drost-Hansen, W., and Clegg, J. (1979). "Cell Associated Water." Academic Press, New York.
- Elul, R. (1974). *Neuroscience Res. Program, Bull.* **12**, 97.
- Fröhlich, F. (1973). In "Cooperative Phenomena" (H. Haken and M. Wagner, eds.), Vol. VII. Springer-Verlag, Berlin and New York.
- Fröhlich, F. (1977) In "Synergetics" (H. Haken, ed.), p. 267. Springer-Verlag, Berlin and New York.
- Fröhlich, H. (1958). "Theory of Dielectrics," 2nd ed. Oxford Univ. Press (Clarendon), Oxford.
- Fröhlich, H. (1968). *Int. J. Quantum Chem.* **2**, 641.
- Fröhlich, H. (1969). In "Theoretical Physics and Biology" (M. Marois, ed.), p. 13. North-Holland Publ., Amsterdam.
- Fröhlich, H. (1970). *Nature (London)* **228**, 1093.
- Fröhlich, H. (1972). *Phys. Lett. A* **39**, 153.
- Fröhlich, H. (1973a). *Riv. Nuovo Cimento* **3**, 490.
- Fröhlich, H. (1973b). *J. Collect. Phenom.* **1**, 101.
- Fröhlich, H. (1975a). *Proc. Natl. Acad. Sci. U.S.A.* **72**, 4211.
- Fröhlich, H. (1975b). *Phys. Lett. A* **51**, 21.
- Fröhlich, H. (1977a). *Riv. Nuovo Cimento* **7**, 399.
- Fröhlich, H. (1977b). *BioSystems* **8**, 193.
- Fröhlich, H. (1977c). *Neurosci. Res. Program, Bull.* **15**, 67.
- Fröhlich, H. (1978). *IEEE Trans. Microwave Theory Tech.* **26**, 613.
- Genzel, L., Keilmann, F., Martin, T. P., Winterling, G., Yacobi, Y., Fröhlich, H., and Makinen, M. W. (1976). *Biopolymers* **15**, 219.
- Genzel, L. (1978). *Phys. Lett. A* **65**, 371.
- Green, D. E. (1974). *Ann. N.Y. Acad. Sci.* **227**, 6.
- Grundler, W., and Keilmann, F. (1978). *Z. Naturforsch., Teil C* **33**, 15.
- Grundler, W., Keilmann, F., and Fröhlich, H. (1977). *Phys. Lett. A* **62**, 463.
- Haken, H. (1973). In "Cooperative Phenomena" (H. Haken and M. Wagner, eds.), p. 363. Springer-Verlag, Berlin and New York.
- Holland, B. (1972). *J. Theor. Biol.* **35**, 395.
- Kaiser, F. (1977a). *Phys. Lett. A* **62**, 63.
- Kaiser, F. (1977b). *Biol. Cybernet.* **27**, 155.
- Kaiser, F. (1978a). *Z. Naturforsch., Teil A* **33**, 294.
- Kaiser, F. (1978b). *Z. Naturforsch., Teil A* **33**, 418.
- Koshland, D. E., and Neet, K. E. (1968). *Annu. Rev. Biochem.* **37**, 359 and 380.
- Kuhn, H. (1949). *J. Chem. Phys.* **17**, 1198.
- Lin, J. C. (1977). *IEEE Trans. Microwave Theory Tech.* **25**, 605.
- Livshits, M. A. (1977). *Biofizika* **22**, 743.
- Luria, E. I. (1975). *Sci. Am. Dec.*, p. 30.
- Manoilov, S. E., Chistyakova, E. N., Kondrateva, V. F., and Strelkova, M. A. (1974). *Sov. Phys. — Usp. (Engl. Transl.)* **16**, 573.
- Mills, R. E. (1979). *Phys. Lett. A* **74**, 278.
- Monod, J. (1972). "Chance and Necessity" (transl.), p. 95. Collins, London.

- Motzkin, S., Birenbaum, L., Rosenthal, S., Rubenstein, C., Davidow, S., Remily, R., and Melnick, R. (1979). *Bioelectromagn. Symp.*, 1979.
- Palma, M. U. (1973). In "From Theoretical Physics to Biology" (M. Marois, ed.), p. 21. Karger, Basel.
- Pethig, R. (1978). *Int. J. Quantum Chem.* **5**, 159.
- Pohl, H. A. (1979). Do cells in the reproductive state exhibit a Fermi-Pasta-Ulam resonance? Research Note 98. Quantum Theoretical Research Group, Oklahoma State University, Stillwater.
- Prohofsky, E. W., and Eyster, J. M. (1974). *Phys. Lett. A* **50**, 329.
- Rose, A. (1970). *Image Technol.* **12**, 13.
- Schmitt, F. O., and Samson, F. E. (1969). *Neurosci. Res. Program, Bull.* **7**, 247.
- Schwan, H. P. (1974). In "Biological Effects and Health Hazards of Microwave Radiation," (P. Czernski, ed.), p. 156. Polish Medical Publishers, Warsaw.
- Sevast'yavona, L. A., and Vilenskaya, R. L. (1974). *Sov. Phys.—Usp. (Engl. Transl.)* **16**, 570.
- Shaya, S. Y., and Smith, C. W. (1977). *J. Collect. Phenom.* **2**, 215.
- Smolyanskaya, A. Z., and Vilenskaya, R. L. (1974). *Trans. Sov. Phys. Usp.* **16**, 571.
- Stuchly, M. A. (1979). *Radiat. Environ. Biophys.* **16**, 1.
- Takashima, S., Onaral, B., and Schwan, H. P. (1979). *Radiat. Environ. Biophys.* **16**, 25.
- Tuengler, P., Keilmann, F., and Genzel, L. (1979). *Z. Naturforsch., Teil C* **34**, 60.
- Vento, G., Palma, M. U., and Indovina, P. (1979). *J. Chem. Phys.* **70**, 2848.
- Watson, J. D. (1970). "Molecular Biology of the Gene," p. 187. Benjamin, New York.
- Webb, S. J. (1975). *Ann. N.Y. Acad. Sci.* **247**, 327.
- Webb, S. J. (1979). *Phys. Lett. A* **73**, 145.
- Webb, S. J. (1980). *Phys. Rep.* (in press).
- Webb, S. J., and Booth, A. D. (1969). *Nature (London)* **222**, 1199.
- Webb, S. J., and Dodds, D. E. (1968). *Nature (London)* **218**, 374.
- Webb, S. J., and Stoneham, M. E. (1977). *Phys. Lett. A* **60**, 267.
- Webb, S. J., Stoneham, M. E., and Fröhlich, H. (1977a). *Phys. Lett. A* **63**, 407.
- Webb, S. J., Lee, K., and Stoneham, M. E. (1977b). *Int. J. Quantum Chem., Quant. Biol. Symp.* **4**, 277.
- Wu, T. M., and Austin, S. (1977). *Phys. Lett. A* **64**, 151.
- Wu, T. M., and Austin, S. (1978a). *J. Theor. Biol.* **71**, 209.
- Wu, T. M., and Austin, S. (1978b). *Phys. Lett. A* **65**, 74.
- Wu, T. M., and Austin, S. (1979). *Phys. Lett. A* **73**, 266.